

Peering into the twilight - a systematic review on visual-based investigation of mesophotic ecosystems - TROPICAL

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Table of Contents

DATA PREPARATION	2
Load libraries	2
Load data.....	3
Subset for studies encompassing tropical latitudinal region	3
Cleaning.....	3
MISSING / ND / NOTAPP VALUES.....	3
GEOGRAPHICAL DISTRIBUTION	5
Provinces.....	5
Ecoregions.....	6
Realms	8
Export geographical distribution summary tables	8
COUNT COLUMNS WITH COMBO	8
Export summary tables.....	9
COUNT METHODS (PURELY TAXONOMIC STUDIES EXCLUDED)	11
COUNT COLUMNS SPLIT.....	14
COUNT METHODS (PURELY TAXONOMIC STUDIES EXCLUDED)	17
PUBLICATIONS PER DEPTH	20
PLATFORM BY DEPTH.....	21
DIRECT - INDIRECT BY DECADE	28
FIELD & INDICATORS	30
Calculate # of ecological indicators per record	35
SAMPLING UNIT.....	36
Prepare a data frame for sampling unit.....	36

Calculate # of sampling unit per record.....	37
Simplified sampling units.....	38
Quadrat extracted from transect	42
DATA TYPE.....	42
HABIT.....	44
SUBSTRATE	46
BENTHIC POSITION	49
TARGET	51
Target level	51
Detected phyla in community studies	53
Target taxonomic groups not at the community level.....	56
New species described per phyla	59
New species descriptions that included molecular analyses	62
DATA CROSSING	63
VISUALLY SURVEYED SURFACE	63
Observation	63
Temp replicates & exp.....	63
Health	64
ROV & sample	64
Molecular techniques in taxonomic and non-taxonomic studies.....	64
SESSION INFORMATION AND REFERENCES.....	65

DATA PREPARATION

Load libraries

```
library(here)

## here() starts at C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/Mesophotic_systematic_lit_rev_analysis

library(dplyr)

##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library(ggplot2)
library(readxl) # to import excel data sets
library(magrittr)
library(stringr) #many data wrangling functions
library(readxl)
```

Load data

```
tropical <- read_excel(here("Extracted_data_analysis", "Input_data", "extracted_raw.xlsx"))
```

Subset for studies encompassing tropical latitudinal region

```
tropical <- subset(tropical, str_detect(tropical$latitudinalRegion, "tropical"))
```

Cleaning

Clean year columns

```
#keep the year only in the year column (in some rows exact time is specified)
tropical$publicationYear <- stringr::str_remove(tropical$publicationYear,
  pattern = "\\-.*")
```

MISSING / ND / NOTAPP VALUES

Create a summary table

```
library(purrr)

##
## Attaching package: 'purrr'

## The following object is masked from 'package:magrittr':
##
##   set_names

library(tibble)

# Get all column names
cols <- names(tropical[, -c(1:13)])

# Create a summary list
summary_list <- map(cols, function(col) {
```

```

data <- tropical[,-c(1:13)][[col]]

tibble(
  Column      = col,
  unique      = paste(unique(data), collapse = ", "),
  nd          = sum(data == "nd", na.rm = TRUE),
  notapp      = sum(data == "notapp", na.rm = TRUE),
  assigned    = sum(!is.na(data) & data != "nd" & data != "notapp"),
  `NA`        = sum(is.na(data)),
  total       = sum(data == "nd", na.rm = TRUE) +
               sum(data == "notapp", na.rm = TRUE) +
               sum(!is.na(data) & data != "nd" & data != "notapp")
)
})

# Combine into one dataframe
summary_df <- bind_rows(summary_list)

summary_df

## # A tibble: 72 × 7
##   Column      unique      nd notapp assigned `NA`
##   <chr>      <chr>      <int> <int>    <int> <int>
##   <int>
## 1 meso_direct_indirect indirect, mixed, di...    3     0    442     0
##   445
## 2 meso_samplingPlatform acou+bruv+tcs, bruv...    3     0    442     0
##   445
## 3 taxonomic_publication NA, taxa, taxa++        0     0     61    384
##   61
## 4 meso_substrate      hard, mixed, nd, so...   19     0    390     36
##   409
## 5 meso_habitat        coralreef, mixed, N...   11     0    418     16
##   429
## 6 meso_protection     mixed, nd, yes, no    346     0     99     0
##   445
## 7 taxonomicTargetLevel comm, group, single...    0     3    442     0
##   445
## 8 habit               mixed, vagile, sess...    0     2    443     0
##   445
## 9 environmentalPosition dem+epi, epi, endo+...    0     1    444     0
##   445
## 10 detected_meso_seagrass NA, x, notapp          0     1     6    438
##   7
## # i 62 more rows

rm(summary_list, cols)

```

GEOGRAPHICAL DISTRIBUTION

```
location <- tropical[,c (
  "recordTitle", "DOI", "publicationYear", "prov_code", "ecoreg_code")]
```

Load Marine Ecoregion of the World data

```
meow <- read.csv(file=here("Extracted_data_analysis", "Input_data", "MEOW.csv"))
```

Provinces

```
prov <- location

prov_lvls <-
  unique(
    unlist(
      str_split(location$prov_code, "\\_")
    )
  )

prov[,prov_lvls] <- sapply(prov_lvls,
  function(y){
    str_count(location$prov_code, y)
  }
)

rm(prov_lvls)
```

Create a column with the sum of provinces per publication

```
prov %<>%
  mutate(
    prov_per_record = rowSums(
      across(
        where(is.numeric)
      )
    )
  )

prov$prov_per_record <- as.factor(prov$prov_per_record)

prov_per_record_df <- prov %>%
  group_by(prov_per_record) %>%
  summarise(n_records = length(prov_per_record)) %>%
  mutate(percentage= round(n_records / sum(n_records) * 100,0))

prov_per_record_df

## # A tibble: 8 × 3
##   prov_per_record n_records percentage
##   <fct>          <int>      <dbl>
```

```
## 1 1          70          16
## 2 2          298         67
## 3 3           60         13
## 4 4           3           1
## 5 5           8           2
## 6 6           4           1
## 7 7           1           0
## 8 21          1           0
```

Create a data frame with number of publication per province

```
prov_tot <-
  prov[, -(length(prov)-1)] %>% # filter out the sum column
  summarise(across(where(is.numeric),
                    list(sum)
                  )
            )

prov_tot <- reshape2::melt(prov_tot, id.vars = NULL)

colnames(prov_tot) <- c("Province", "Count")

prov_tot %<>%
  mutate_at("Province",
            stringr::str_replace, "_1", "")
```

Ecoregions

```
ecoreg <- location
ecoreg_lvls <-
  unique(
    unlist(
      str_split(ecoreg$ecoreg_code, "\\_")
    )
  )

ecoreg_lvls <- ecoreg_lvls[! ecoreg_lvls %in% c("")]

ecoreg[, ecoreg_lvls] <- sapply(ecoreg_lvls,
                                function(y){
                                  str_count(ecoreg$ecoreg_code, y)
                                }
                                )
```

Create a column with the number of ecoregions per study

```
ecoreg %<>%
  mutate(
    ecoreg_per_record = rowSums(
      across(
```

```

    where(is.numeric)
  )
)
)

```

```
ecoreg$ecoreg_per_record <- as.factor(ecoreg$ecoreg_per_record)
```

Calculate number of records by number of ecoregions in %

```

ecoreg_per_record_df <- ecoreg %>%
  group_by(ecoreg_per_record) %>%
  summarise(n_records = length(ecoreg_per_record)) %>%
  mutate(percentage= round(n_records / sum(n_records) * 100,1))

```

```
ecoreg_per_record_df
```

```

## # A tibble: 6 × 3
##   ecoreg_per_record n_records percentage
##   <fct>            <int>      <dbl>
## 1 1                382        85.8
## 2 2                 39         8.8
## 3 3                 18         4
## 4 4                  3         0.7
## 5 5                  2         0.4
## 6 12                 1         0.2

```

Create a data frame with number of publication per ecoregion

```

ecoreg_tot <-
  ecoreg [, -(length(ecoreg)-1)] %>% # filter out the sum column
  summarise(across(where(is.numeric),
                    list(sum)
                  )
            )

```

```
ecoreg_tot <- reshape2::melt(ecoreg_tot, id.vars = NULL)
```

```
colnames(ecoreg_tot) <- c("Ecoregion", "Count")
```

```

ecoreg_tot %<>%
  mutate_at("Ecoregion",
            stringr::str_replace, "_1", "")

```

Add count column to Meow and set 0 to all the ecoregions where no study was conducted

```

colnames(ecoreg_tot)[colnames(ecoreg_tot) == 'Ecoregion'] <- 'ECO_CODE'
ecoreg_tot$ECO_CODE <- as.character(ecoreg_tot$ECO_CODE)
meow$ECO_CODE <- as.character(meow$ECO_CODE)

```

```
ecoreg_tot <-
  left_join(meow, ecoreg_tot, by = 'ECO_CODE') %>%
  mutate(Count = replace(Count, is.na(Count), 0))
```

Export count per ecoregion to csv for the map in QGIS

```
writexl::write_xlsx(ecoreg_tot,
  here("Extracted_data_analysis", "tropical", "Output-Transformed_data",
    "eco_reg_tot.xlsx"))
```

Realms

```
realms <- ecoreg_tot %>%
  group_by(RLM_CODE, REALM) %>%
  summarise(Count=sum(Count))

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by RLM_CODE and REALM.
## i Output is grouped by RLM_CODE.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(RLM_CODE, REALM))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

realms %<>%
  rename(Code = RLM_CODE,
    Realm = REALM)

writexl::write_xlsx(realms,
  here("Extracted_data_analysis", "tropical", "Output-Transformed_data",
    "Realm_count.xlsx"))
```

Export geographical distribution summary tables

```
writexl::write_xlsx(list(realms, prov_tot, prov_per_record_df, ecoreg_tot, ec
oreg_per_record_df),
  here("Extracted_data_analysis", "tropical", "Output-Transformed_data",
    "geographical_distribution_summary.xlsx"))
```

COUNT COLUMNS WITH COMBO

Create a subset of columns

```
count_columns <- tropical %>%
  select(File,
    meso_substrate,
    habit,
    environmentalPosition,
    observationalORexperimental,
    taxonomicTargetLevel,
    indicators,
```



```
collectedDataType,  
molecularAnalyses)
```

Prepare data set with columns that need count only and add percentage column

```
count_columns[,] <- lapply(count_columns[,], as.factor)  
  
count_results <- list()  
  
for (col_name in colnames(count_columns)) {  
  counts <- count(count_columns, !!sym(col_name))  
  
  count_results[[col_name]] <- counts  
  
  colnames(count_results[[col_name]]) <- c(col_name, "Count")  
  
  count_results[[col_name]] %<>%  
    filter(  
      if_all(.cols = everything(), ~ !grepl("notapp", .x))  
    ) %>%  
    # delete rows with "notapp"  
    tidyr::drop_na()  
  
  count_results[[col_name]] %<>%  
    mutate(Percentage= round(Count / sum(Count) * 100,2))  
}  
  
rm(col_name, counts)  
  
count_results <- count_results[names(count_results) != "File"]
```

Export summary tables

```
library(openxlsx)  
  
write.xlsx(count_results,  
           file = here("Extracted_data_analysis",  
                       "tropical",  
                       "Output-Transformed_data",  
                       "summary_counts_question_1-3.xlsx"))
```

Create treemaps charts

```
treemap_func = function(dat,var) {  
  ggplot(dat,  
    aes(area=Count,  
        fill=.data[[var]],  
        line = "black",  
        label=paste0(.data[[var]],  
                      " \n ",Percentage,"%"))
```

```

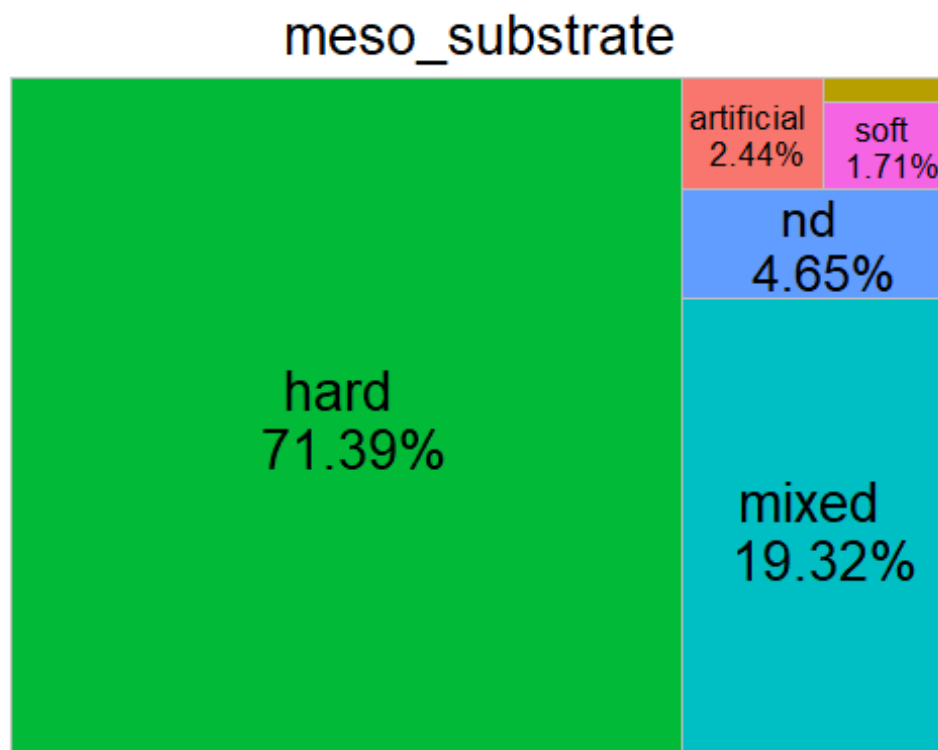
)+
treemapify::geom_treemap(layout = "squarified")+
treemapify::geom_treemap_text(place = "centre",
                             size = 20)+
labs(title = var)+
theme(plot.title = element_text(hjust = 0.5, size= 20),
      legend.position = "none")
}

treemap_counts <- list()

for (col_name in colnames(count_columns[, -1])) {
  treemap_counts[[col_name]] <- treemap_func(dat= count_results[[col_name]],
var = col_name)
}

treemap_counts[[1]]

```



Save all tree maps plots

```

lapply(names(treemap_counts), function(x){
  ggsave(filename=paste0(x, ".tiff"),
          plot=treemap_counts[[x]],

```

```

    units = "in", width=12, height=4, dpi=300, compression = 'lzw',
    path = here("Extracted_data_analysis",
                "tropical",
                "Output-Visualization",
                "Treemap_counts"),
    create.dir = T
  )
})

lapply(names(treemap_counts), function(x){
  ggsave(filename=paste0(x, ".png"),
    plot=treemap_counts[[x]],
    units = "in", width=12, height=4, dpi=300,
    path = here("Extracted_data_analysis",
                "tropical",
                "Output-Visualization",
                "Treemap_counts"),
    create.dir = T
  )
})

```

COUNT METHODS (PURELY TAXONOMIC STUDIES EXCLUDED)

Remove purely taxonomic studies and subset columns (taxa++ was assigned to studies that led to the description of new species but where also ecological e.g. describing community for a given taxonomic group)

```

methods <- tropical %>%
  filter(taxonomic_publication == "taxa++" | is.na(taxonomic_publication) )
  %>%
  select(File,
    meso_design,
    meso_replicates,
    temporalReplicates,
    abioticVariablesCollection,
    abioticVariables)

count_columns_method <- methods

```

Prepare data set with columns that need count only and add percentage column

```

count_columns_method[, ] <- lapply(count_columns_method[, ], as.factor)

count_results_method <- list()

for (col_name in colnames(count_columns_method)) {
  counts <- count(count_columns_method, !!sym(col_name))
}

```

```

count_results_method[[col_name]] <- counts

colnames(count_results_method[[col_name]]) <- c(col_name, "Count")

count_results_method[[col_name]] %<>%
  filter(
    if_all(.cols = everything(), ~ !grepl("notapp", .x))
  ) %>%
  # delete rows with "notapp"
  tidyr::drop_na()

count_results_method[[col_name]] %<>%
  mutate(Percentage= round(Count / sum(Count) * 100,1))
}

rm(col_name, counts)

count_results_method <- count_results_method[names(count_results_method) != "
File"]

```

Export summary tables

```

library(openxlsx)

write.xlsx(count_results_method,
           file = here("Extracted_data_analysis", "Output-Transformed_data", "summary_counts_question_4.xlsx"))

```

Create treemaps charts

```

treemap_func = function(dat,var) {
  ggplot(dat,
    aes(area=Count,
        fill=.data[[var]],
        line = "black",
        label=paste0(.data[[var]],
                     " \n ",Percentage,"%"))
  )+
  treemapify::geom_treemap(layout = "squarified")+
  treemapify::geom_treemap_text(place = "centre",
    size = 20)+
  labs(title = var)+
  theme(plot.title = element_text(hjust = 0.5, size= 20),
    legend.position = "none")
}

```

```

treemap_counts <- list()

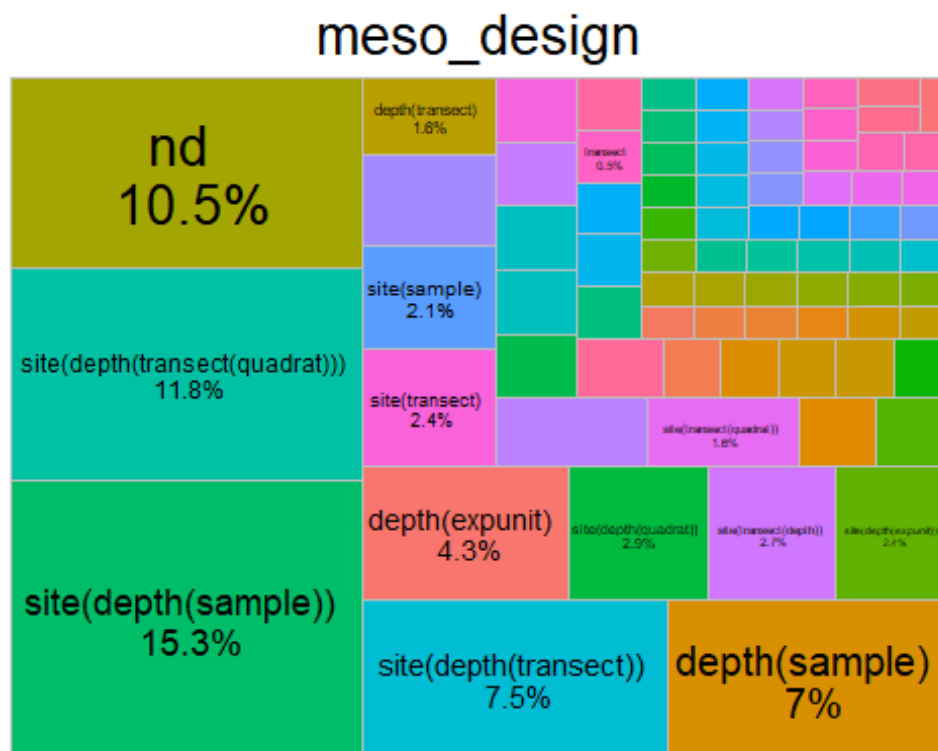
for (col_name in colnames(count_columns_method[, -1])) {

  treemap_counts[[col_name]] <- treemap_func(dat= count_results_method[[col_name]], var = col_name)

}

treemap_counts[[1]]

```



Save all tree maps plots

```

lapply(names(treemap_counts), function(x){
  ggsave(filename=paste0("treemap",x,".tiff"),
    plot=treemap_counts[[x]],
    units = "in", width=12, height=4, dpi=300, compression = 'lzw',
    path = here("Extracted_data_analysis","tropical",
      "Output-Visualization","Treemaps_methods"),
    create.dir = T
  )
})

lapply(names(treemap_counts), function(x){
  ggsave(filename=paste0("treemap",x,".png"),
    plot=treemap_counts[[x]],

```

```

    units = "in", width=12, height=4, dpi=300,
    path = here("Extracted_data_analysis", "tropical", "Output-Visualization", "Treemaps_methods"),
    cal",
    create.dir = T
  )
})

```

COUNT COLUMNS SPLIT

Create a list of data frame splitting values

```

split_columns <- tropical %>%
  select(meso_substrate,
         habit,
         environmentalPosition,
         observationalORexperimental,
         taxonomicTargetLevel,
         indicators,
         collectedDataType,
         molecularAnalyses)

```

```
split_columns[,] <- lapply(split_columns[,], as.factor)
```

Prepare data set with columns

```

split_results <- list()

for (col_name in colnames(split_columns)) {
  # Split the values of each column by "+"
  lvls <- unique(unlist(str_split(split_columns[[col_name]], "\\+")))

  # Initialize a list to store counts for each level
  split_counts <- sapply(lvls, function(y) {
    # Count how many times each unique level appears in the column
    str_count(split_columns[[col_name]], fixed(y))
  })

  split_results[[col_name]] <- split_counts
}

## Warning in stri_count_fixed(string, pattern, opts_fixed = opts(pattern)):
## empty
## search patterns are not supported

rm(lvls)

```

Add percentage column

```

split_tot <- list()

for(col_name in colnames(split_columns)) {
  split_results[[col_name]] <- as.data.frame(split_results[[col_name]])

  split_results[[col_name]] %<>% # delete rows with missing values
    select(where(function(x) any(!is.na(x)))) %>%
    select(-any_of("notapp")) %>%
    tidyr::drop_na()

  split_tot[[col_name]] <- as.data.frame(split_results[[col_name]]) %>%
    summarise(across(where(is.numeric), list(sum)))

  split_tot[[col_name]] <- reshape2::melt(split_tot[[col_name]], id.vars = N
ULL)

  colnames(split_tot[[col_name]]) <- c(col_name, "Count")

  split_tot[[col_name]] %<>%
    mutate_at(col_name,
              stringr::str_replace, "_1", "")

  split_tot[[col_name]] %<>%
    mutate(Percentage= round(Count / nrow(split_results[[col_name]] ) * 100,
2))
}

rm(col_name)

```

Export summary tables

```

library(openxlsx)

write.xlsx(split_tot,
           file = here("Extracted_data_analysis", "tro
pical", "Output-Transformed_data",
"summary_split_counts_question_1_3.xlsx"))

```

Create bar charts for each collected parameter

```

barchart_func = function(dat,var) {

  ggplot(dat,
         aes(x = reorder(.data[[var]], Count), y = Count, fill= .data[[var]]))
+
  geom_bar(stat = "identity", fill = "#0097A7") +
  geom_text(

```

```

data = dat %>% filter(Percentage >= 2),
aes(label = paste0(round(Percentage, 0), "%"),
position = position_stack(vjust =0.5),
color = "black",
size = 2.5
)+
labs(y = "# Publications", x = var) +
theme_minimal() +
coord_flip()+ # optional: flips for horizontal bars
theme(legend.position = "none")
}

barchart_counts <- list()

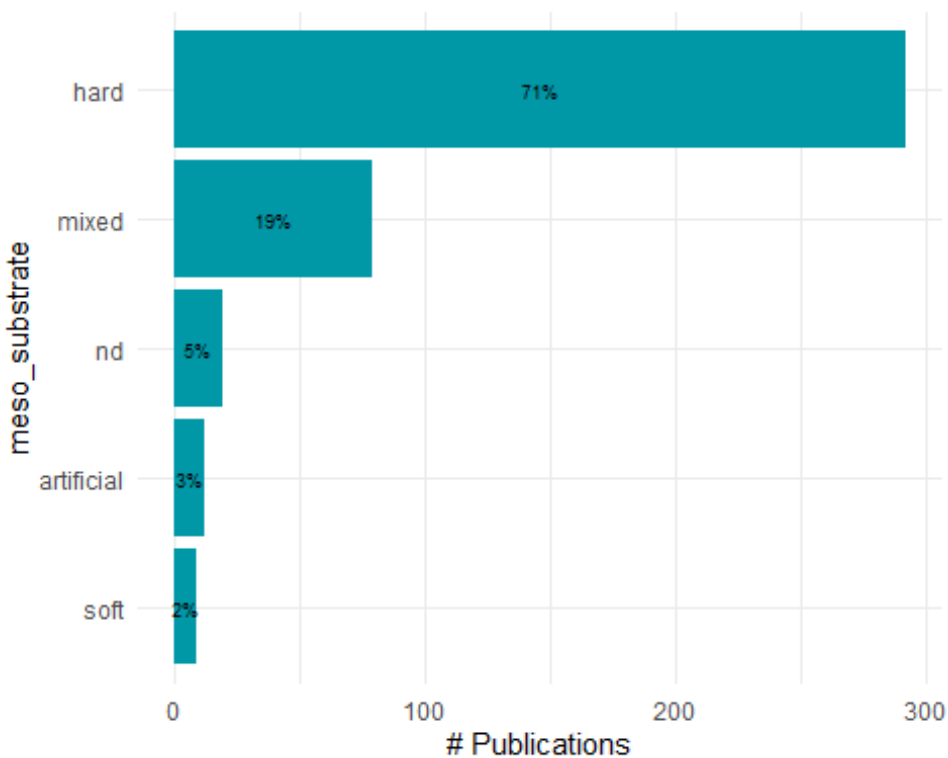
for (col_name in colnames(split_columns)) {

  barchart_counts[[col_name]] <- barchart_func(dat= split_tot[[col_name]], va
r = as.character(col_name))

}

barchart_counts[[1]]

```



```

lapply(names(barchart_counts), function(x){
  ggsave(filename=paste0("bar_",x,".tiff"),

```



```

        plot=barchart_counts[[x]],
        units = "in", width=12, height=4, dpi=300, compression = 'lzw',
        path = here("Extracted_data_analysis", "tropi
cal", "Output-Visualization", "Barcharts_split"),
        create.dir = T
    )
})

lapply(names(barchart_counts), function(x){
  ggsave(filename=paste0("bar_",x,".png"),
    plot=barchart_counts[[x]],
    units = "in", width=12, height=4, dpi=300,
    path = here("Extracted_data_analysis", "tropi
cal", "Output-Visualization", "Barcharts_split"),
    create.dir = T
  )
})

```

COUNT METHODS (PURELY TAXONOMIC STUDIES EXCLUDED)

Create data frame with subset

```

split_columns_methods <- methods

split_columns_methods[,] <- lapply(split_columns_methods[,], as.factor)

```

Prepare data set with columns

```

split_results_methods <- list()

for (col_name in colnames(split_columns_methods)) {
  # Split the values of each column by "+"
  lvls <- unique(unlist(str_split(split_columns_methods[[col_name]], "\\+")))

  # Initialize a list to store counts for each level
  split_counts_methods <- sapply(lvls, function(y) {
    # Count how many times each unique level appears in the column
    str_count(split_columns_methods[[col_name]], fixed(y))
  })

  split_results_methods[[col_name]] <- split_counts_methods
}

rm(lvls)

```

Add percentage column

```

split_tot_methods <- list()

for(col_name in colnames(split_columns_methods)) {
  split_results_methods[[col_name]] <- as.data.frame(split_results_methods[[col_name]])

  split_results_methods[[col_name]] %<>% # delete rows with missing values
    select(where(function(x) any(!is.na(x)))) %>%
    select(-any_of("notapp")) %>%
    tidyr::drop_na()

  split_tot_methods[[col_name]] <- as.data.frame(split_results_methods[[col_name]]) %>%
    summarise(across(where(is.numeric), list(sum)))

  split_tot_methods[[col_name]] <- reshape2::melt(split_tot_methods[[col_name]], id.vars = NULL)

  colnames(split_tot_methods[[col_name]]) <- c(col_name, "Count")

  split_tot_methods[[col_name]] %<>%
    mutate_at(col_name,
              stringr::str_replace, "_1", "")

  split_tot_methods[[col_name]] %<>%
    mutate(Percentage= round((Count / nrow(split_results_methods[[col_name]])) * 100,2))
}

rm(col_name)

```

Export summary tables

```

library(openxlsx)

write.xlsx(split_results_methods,
           file = here("Extracted_data_analysis", "Output-Transformed_data", "summary_split_counts_question_4.xlsx"),
           pical, "tro")

```

Create bar charts per each parameter collected

```

barchart_func = function(dat,var) {

  ggplot(dat,
         aes(x = reorder(.data[[var]], Count), y = Count, fill= .data[[var]]))
+

```

```

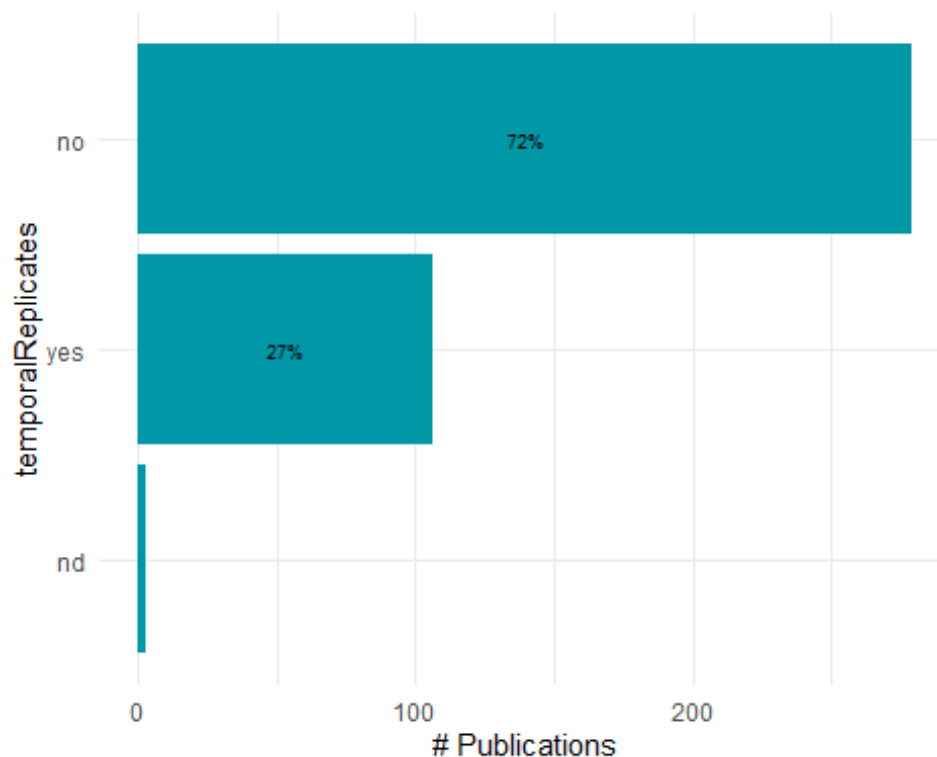
geom_bar(stat = "identity", fill = "#0097A7") +
geom_text(
  data = dat %>% filter(Percentage >= 2),
  aes(label = paste0(round(Percentage, 0), "%"),
    position = position_stack(vjust = 0.5),
    color = "black",
    size = 2.5
  )+
labs(y = "# Publications", x = var) +
theme_minimal() +
coord_flip()+ # optional: flips for horizontal bars
theme(legend.position = "none")
}

barchart_split_methods <- list()

for (col_name in colnames(split_columns_methods)) {
  barchart_split_methods[[col_name]] <- barchart_func(dat= split_tot_methods
[[col_name]], var = as.character(col_name))
}

barchart_split_methods[[4]]

```



Export bar charts in a dedicated folder

```
lapply(names(barchart_counts), function(x){
  ggsave(filename=paste0("bar_",x,".tiff"),
    plot=barchart_counts[[x]],
    units = "in", width=12, height=4, dpi=300, compression = 'lzw',
    path = here("Extracted_data_analysis", "tropi
cal", "Output-Visualization", "Barcharts_split_methods
"), create.dir = T
  })

lapply(names(barchart_counts), function(x){
  ggsave(filename=paste0("bar_",x,".png"),
    plot=barchart_counts[[x]],
    units = "in", width=12, height=4, dpi=300,
    path = here("Extracted_data_analysis", "tropi
cal", "Output-Visualization", "Barcharts_split"), creat
e.dir = T
  })
```

PUBLICATIONS PER DEPTH

```
tropical$minimumSurveyedDepthInMeters <- as.numeric(tropical$minimumSurveyedD
epthInMeters)
tropical$maximumSurveyedDepthInMeters <- as.numeric(tropical$maximumSurveyedD
epthInMeters)

# Define depth bins
bins <- seq(31, 151, by = 10)

# Create a data frame to hold bin counts
pubs_depth <- data.frame(
  bin_start = bins[-length(bins)],
  bin_end = bins[-1]-1,
  count = 0
)

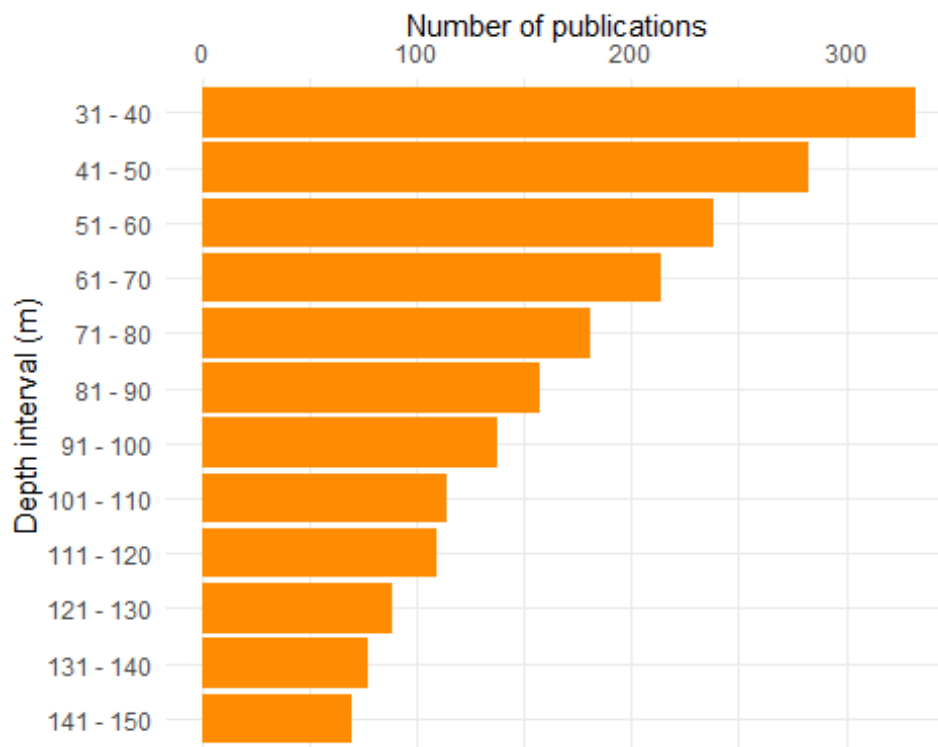
# For each bin, count how many studies cover that depth range
for (i in 1:nrow(pubs_depth)) {
  bin_start <- pubs_depth$bin_start[i]
  bin_end <- pubs_depth$bin_end[i]

  pubs_depth$count[i] <- sum(tropical$maximumSurveyedDepthInMeters >= bin_sta
rt & tropical$minimumSurveyedDepthInMeters < bin_end, na.rm = TRUE)
}

pubs_depth$depth_bin <- paste(pubs_depth$bin_start, "-", pubs_depth$bin_end)
```

```
pubs_depth$depth_bin <- reorder(pubs_depth$depth_bin, -pubs_depth$bin_start)
```

```
ggplot(pubs_depth, aes(x = factor(depth_bin), y = count)) +  
  geom_bar(stat = "identity", fill = "darkorange") +  
  labs(x = "Depth interval (m)", y = "Number of publications") +  
  theme_minimal()+  
  coord_flip()+  
  scale_y_continuous(position="right")
```



Export pubs per depth bar plot

PLATFORM BY DEPTH

Create a platform by depth data frame and keep meso_samplingPlatform

```
platform_depth <- tropical %>%  
  select(File,  
    mininumSurveyedDepthInMeters,  
    maximumSurveyedDepthInMeters,  
    meso_direct_indirect,  
    meso_samplingPlatform,  
    max_depth_hov, min_depth_hov,  
    max_depth_rov, min_depth_rov,
```

```

max_depth_ccr, min_depth_ccr,
max_depth_open, min_depth_open,
max_depth_scuba, min_depth_scuba
)

```

Split the platform column in one binary column per platform

```

platform_lvls <-
  unique(
    unlist(
      str_split(platform_depth$meso_samplingsPlatform, "\\+")
    )
  )

platform_lvls <- platform_lvls[! platform_lvls %in% c("", NA, "notapp", ",", "nd")]

platform_depth[, platform_lvls] <- sapply(platform_lvls,
  function(y){
    str_count(platform_depth$meso_samplingsPlatform, y)
  })

rm(platform_lvls)

platform_depth[, -c(1:15)] <- lapply(platform_depth[, -c(1:15)], as.numeric)
platform_depth[, c(1:15)] <- lapply(platform_depth[, c(1:15)], as.character)

platform_depth %<>%
  rename(open_direct = open,
    ccr_direct = ccr,
    scuba_direct = scuba) %>%
  select(-meso_samplingsPlatform)

```

Create a summary table with # studies per platform

```

platform_summary <- platform_depth %>%
  summarise(across(where(is.numeric), ~ sum(.x, na.rm = TRUE)))

platform_summary <- as.data.frame(t(platform_summary))

platform_summary %<>%
  mutate(Platform = rownames(platform_summary)) %>%
  rename(Count = V1) %>%
  relocate(Count, .after = where(is.character))

platform_summary

```

```
##           Platform Count
## acou      acou      27
## bruv      bruv       8
## tcs       tcs      16
## dredge    dredge    15
## fishery   fishery    6
## scuba_direct scuba_direct 75
## trawl     trawl      2
## open_direct open_direct 168
## hov       hov       86
## rov       rov      101
## ccr_direct ccr_direct  80
## auv       auv        9
## grab      grab      13
## vams      vams        2
## mixed     mixed       1
## drop      drop        8
## station   station     1
```

Create a column with the number of platforms per study

```
platform_per_record <- platform_depth %>%
  filter(meso_direct_indirect != "nd") %>%
  mutate(
    platform_per_record = rowSums(
      across(
        where(is.numeric)
      )
    )
  )

platform_per_record$platform_per_record <- as.factor(platform_per_record$platform_per_record)
```

Calculate number of records by number of platforms in %

```
platform_per_record <- platform_per_record %>%
  group_by(platform_per_record) %>%
  summarise(n_records = length(platform_per_record)) %>%
  mutate(percentage= round(n_records / sum(n_records) * 100,1))
```

platform_per_record

```
## # A tibble: 4 × 3
##   platform_per_record n_records percentage
##   <fct>              <int>         <dbl>
## 1 1                  309          69.9
## 2 2                   98          22.2
## 3 3                   27           6.1
## 4 4                    8           1.8
```

```
platform_depth_mod <- cbind(platform_depth[,c(1:14)],
                             platform_depth$ccr_direct,
                             platform_depth$open_direct,
                             platform_depth$scuba_direct,
                             platform_depth$hov,
                             platform_depth$rov)

names(platform_depth_mod) <- gsub("platform_depth\\$", "", names(platform_depth_mod))
```

```
platform_depth_long <- within(platform_depth_mod, rm(mininumSurveyedDepthInMe
ters, maximumSurveyedDepthInMeters, max_depth_direct, meso_direct_indirect, h
ov, rov, open_direct, ccr_direct, scuba_direct))
```

```
str(platform_depth_long)
```

```
## 'data.frame':    445 obs. of  11 variables:
## $ File           : chr  "AbdulWahabEtAl_2018_Biodiversity_spatial_pattern
s.md" "AbesamisEtAl_2018_Benthic_habitat_fish.md" "AlbanoEtAl_2011_challenge_
small_rare.md" "AlbeldaEtAl_2020_Juvenile_scleractinian_assemblage.md" ...
## $ max_depth_hov  : num  NA NA NA NA NA NA NA NA NA NA 150 ...
## $ min_depth_hov  : num  NA NA NA NA NA NA NA NA NA NA 45 ...
## $ max_depth_rov   : num  NA NA NA NA NA NA 100 70 NA NA ...
## $ min_depth_rov   : num  NA NA NA NA NA NA 20 20 NA NA ...
## $ max_depth_ccr   : num  NA NA NA NA NA NA NA NA NA NA ...
## $ min_depth_ccr   : num  NA NA NA NA NA NA NA NA NA NA ...
## $ max_depth_open  : num  NA NA NA 40 NA NA 53 NA NA NA ...
## $ min_depth_open  : num  NA NA NA 3 NA NA 10 20 NA NA ...
## $ max_depth_scuba : num  NA NA NA NA NA 55 NA NA NA NA ...
## $ min depth scuba : num  NA NA 1 NA 72 4 NA NA NA NA ...
```

[illegible]


```
mutate(max_depth_open = case_when(max_depth_open >= 150 ~ 150,
                                   TRUE ~ max_depth_open))
```

Set min depth of studies with min depth <30 to 30 so they fit into the plot

```
platform_depth_long %<>%
  mutate(min_depth_scuba = case_when(min_depth_scuba <= 30 ~ 30,
                                     TRUE ~ min_depth_scuba)
  ) %>%
  mutate(min_depth_hov = case_when(min_depth_hov <= 30 ~ 30,
                                   TRUE ~ min_depth_hov)
  ) %>%
  mutate(min_depth_rov = case_when(min_depth_rov <= 30 ~ 30,
                                   TRUE ~ min_depth_rov)
  ) %>%
  mutate(min_depth_ccr = case_when(min_depth_ccr <= 30 ~ 30,
                                   TRUE ~ min_depth_ccr)) %>%
  mutate(min_depth_open = case_when(min_depth_open <= 30 ~ 30,
                                   TRUE ~ min_depth_open))
```

Convert platform_depth_long to long format

```
platform_min_depth_long <- platform_depth_long %>%
  select(-c(max_depth_hov, max_depth_rov, max_depth_open, max_depth_scuba, ma
x_depth_ccr))
```

```
platform_min_depth_long[,c(1,2)] <- lapply(platform_min_depth_long[,c(1,2)],
as.factor)
```

```
platform_min_depth_long %<>%
  reshape2::melt(platform_min_depth_long,
                 id.vars = c("File"),
                 measure.vars = grep("^min_depth_|^max_depth_",
                                     names(platform_min_depth_long),
                                     value = TRUE),
                 variable.name = "Platform",
                 value.name = "min_depth"
  )
```

```
## Warning: attributes are not identical across measure variables; they will
be
## dropped
```

```
platform_max_depth_long <- platform_depth_long %>%
  select(-c(min_depth_hov, min_depth_rov, min_depth_open, min_depth_scuba,
min_depth_ccr))
```

```
platform_max_depth_long %<>%
  reshape2::melt(platform_max_depth_long,
```

```

id.vars = c("File"),
measure.vars = grep("^min_depth_|^max_depth_",
                    names(platform_max_depth_long),
                    value = TRUE),
variable.name = "Platform",
value.name = "max_depth"
)

```

```
platform_depth_lat_long <- cbind(platform_min_depth_long, platform_max_depth_
long$max_depth)
```

```
rm(platform_min_depth_long, platform_max_depth_long)
```

```
platform_depth_lat_long %<>%
  rename(max_depth = `platform_max_depth_long$max_depth`)
```

```
platform_depth_lat_long$Platform <- gsub("min_depth_", "", platform_depth_lat
_long$Platform)
```

```
platform_depth_lat_long$Platform <- gsub("max_depth_", "", platform_depth_lat
_long$Platform)
```

Set proper labelling for platform

```
platform_depth_lat_long %<>%
  mutate(Platform = case_when(Platform == "ccr" ~ "CCR",
                              Platform == "scuba" ~ "SCUBA",
                              Platform == "open" ~ "OC",
                              Platform == "rov" ~ "ROV",
                              Platform == "hov" ~ "HOV",
                              TRUE ~ Platform)
  ) %>%
  mutate(Platform = as.factor(Platform))
```

Add visual column to reorder platform

```
platform_depth_lat_long %<>%
  mutate(Visual = as.factor(case_when(Platform == "OC" |
                                      Platform == "SCUBA" |
                                      Platform == "CCR" ~ "direct",
                                      TRUE ~ "indirect" )
  )
)
```

```
platform_depth_lat_long %<>%
  mutate(Platform = forcats::fct_reorder(Platform, as.integer(Visual)))
```

Create a bar plot with depth bins by platform and latitudinalRegion

```
depth_bins <- tibble(
  bin_start = seq(-30, -140, by = -10),
```

```

    bin_end    = seq(-40, -150, by = -10)
  ) %>%
    mutate(bin_label = paste0(abs(bin_start), "-", abs(bin_end)))

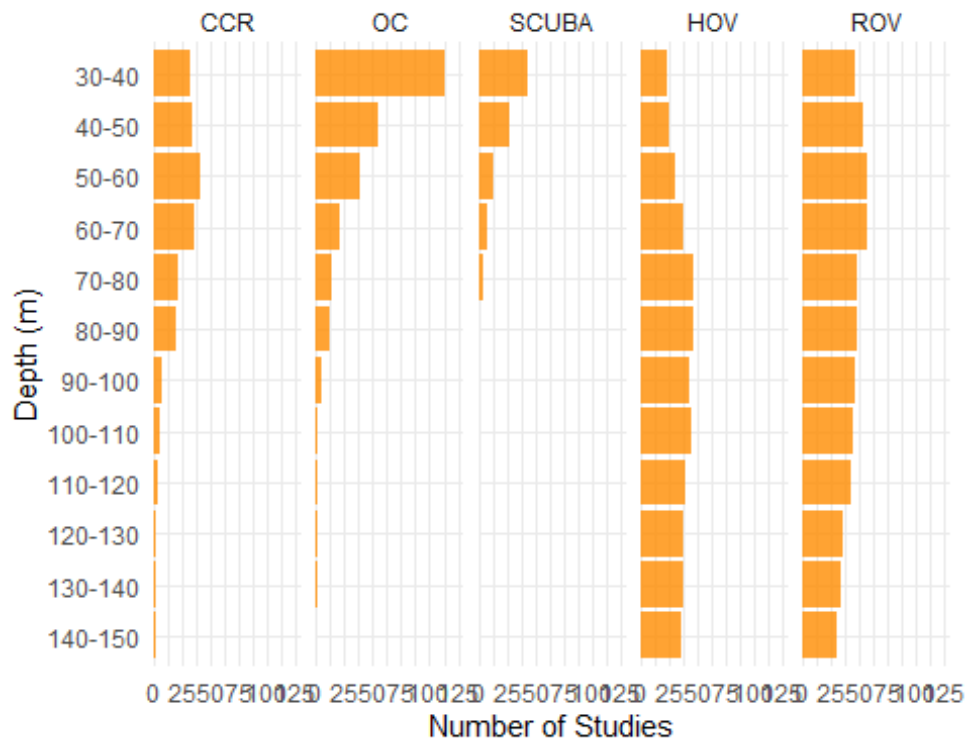
platform_depth_lat_long <- platform_depth_lat_long %>%
  mutate(
    min_depth = as.numeric(min_depth), # positive e.g., 30
    max_depth = as.numeric(max_depth)  # positive e.g., 150
  )

expanded_data <- platform_depth_lat_long %>%
  tidyr::crossing(depth_bins) %>%
  # Convert study range to negative depths: [-max_depth, -min_depth]
  filter(bin_end >= -max_depth, bin_start <= -min_depth) %>%
  mutate(
    bin_label = factor(bin_label, levels = rev(depth_bins$bin_label))
  )

# Plot with count labels on the right
platform_depth_plot <- ggplot(expanded_data, aes(x = bin_label)) +
  geom_bar(position = "stack", alpha= 0.8, fill= "darkorange") +
  scale_y_continuous(name = "Number of Studies",
    expand = expansion(mult = c(0, 0.15))) + # Make room for
r text
  scale_x_discrete(name = "Depth (m)") +
  coord_flip() + # Depth on y-axis (shallow at top)
  facet_wrap(~ Platform, nrow = 1) +
  theme_minimal() +
  theme(legend.position = "none")

platform_depth_plot

```



Export bar plot of most common platform per depth interval

```
ggsave("platform_depth_plot.tiff",
  plot = platform_depth_plot,
  units = "in", width = 8, height = 6, dpi = 300, compression = 'lzw',
  path = here ("Extracted_data_analysis", "tropical",
    "Output-Visualization"))

ggsave("platform_depth_plot.png",
  plot = platform_depth_plot,
  units = "in", width = 8, height = 5,
  path = here ("Extracted_data_analysis", "tropical",
    "Output-Visualization"))
```

DIRECT - INDIRECT BY DECADE

```
dir_ind_by_decade <- tropical

dir_ind_by_decade %<>%
  mutate(decade = case_when(publicationYear >= 1960 & publicationYear
    < 1970 ~ "1970-1979",
    publicationYear >= 1970 & publicationYear <
    1980 ~ "1970-1979",
    publicationYear >= 1980 & publicationYear <
    1990 ~ "1980-1989",
    publicationYear >= 1990 & publicationYear <
```

```

2000 ~ "1990-1999",
                                publicationYear >= 2000 & publicationYear <
2010 ~ "2000-2009",
                                publicationYear >= 2010 & publicationYear <
2020 ~ "2010-2019",
                                publicationYear >= 2020 ~ "2020-2023")
    )

dir_ind_by_decade_lat <- dir_ind_by_decade

dir_ind_by_decade <- dir_ind_by_decade [, c("meso_direct_indirect", "decade")]
%>%
  group_by(decade, meso_direct_indirect) %>%
  tally()

colnames(dir_ind_by_decade) <- c("Decade", "Technique", "Count")

unique(dir_ind_by_decade$Technique)

## [1] "direct"    "mixed"     "indirect"  "nd"

dir_ind_by_decade %<>%
  filter(Technique != "nd" & Technique != "NA") %>% #remove nd from viz becau
se they accounted for a really small percentage
  group_by(Decade) %>%
  mutate(Percentage = round(Count / sum(Count) * 100, 1))

```

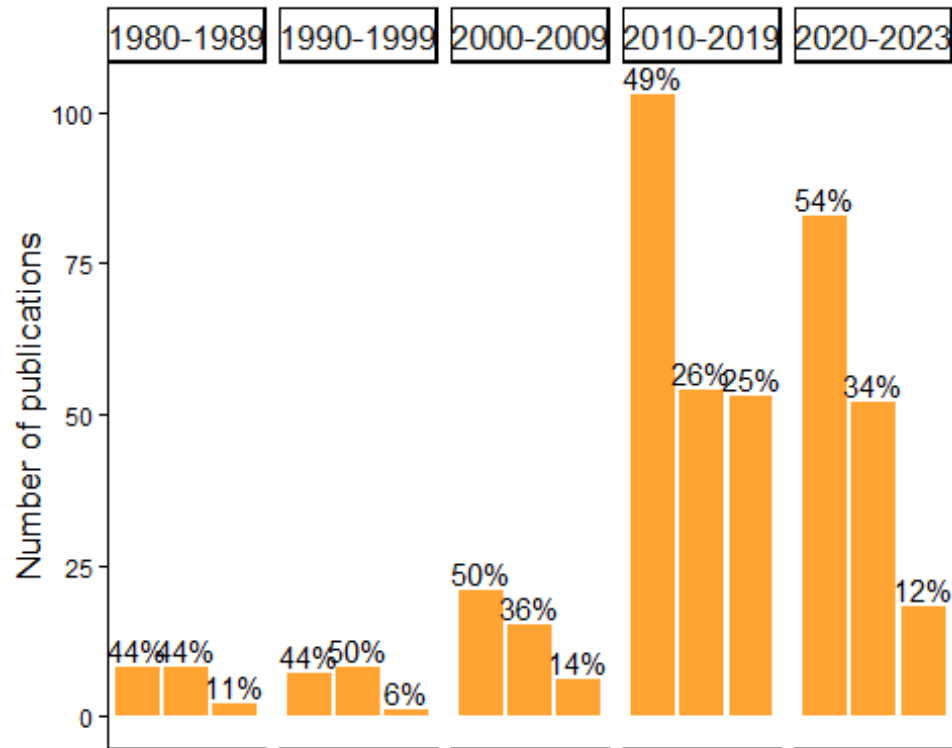
Make bar plot of technique by decade

```

technique_decade_barplot <- ggplot(subset(dir_ind_by_decade,
  dir_ind_by_decade$Decade != "1970-1979"),
  aes(x=Technique, y=Count, label = Technique)
) +
  geom_bar(position="dodge",
    stat="identity", alpha= 0.8, fill = "darkorange" ) +
  geom_text(aes(label = paste(round(Percentage, 0), "%", sep="")),
    parse = F,
    vjust = -0.25,
    size=3.9)+
  facet_grid(~ Decade,
    switch = "y") +
  theme_classic() +
  theme(legend.position = "bottom",
    legend.text = element_text(size = 10),
    axis.text.x = element_blank(),
    axis.text.y = element_text(size=8.7),
    axis.title.x = element_blank(),
    axis.title.y = element_text(size=13),
    strip.text = element_text(size = 13))+
  labs(y= "Number of publications")

```

technique_decade_barplot



Export bar plot of technique by decade

```
ggsave("technique_decade_barplot.tiff",
  plot = technique_decade_barplot,
  units = "in", width = 7, height = 6, dpi = 300, compression = 'lzw',
  path = here ("Extracted_data_analysis", "tropic
al", "Output-Visualization"))

ggsave("technique_decade_barplot.png",
  plot = technique_decade_barplot,
  units = "in", width = 7, height = 6,
  path = here ("Extracted_data_analysis", "tropic
al", "Output-Visualization"))
```

FIELD & INDICATORS

Create subset for field & indicators

```
field_merged <- tropical[,c(59:length(tropical))]

field_merged[,] <- lapply(field_merged[,], as.numeric)

colSums(field_merged)
```

```
##           field_cons           field_commdesc
##           24             147
##           field_map           field_impass
##           46             67
##           field_troph           field_taxo
##           37             107
##           field_dist           field_aqua
##           170             1
##           field_method           field_physio
##           13             124
##           field_func           field_habtype
##           17             0
##           field_fishery           field_stock
##           8             7
##           field_paleo           field_phylo
##           0             31
##           field_rest           field_archaeo
##           1             0
##           field_micro           field_biomedicine
##           7             1
##           field_chem           field_Molecular_ecology
##           1             121
##           field_Acoustics field_environmental_drivers
##           26             137
##           field_Behaviour           field_Geomorphology
##           27             13
##           field_Taxonomy
##           92
```

Select only the most common fields

```
field_merged <- field_merged[, colSums(field_merged) > 30]
```

```
field_merged$field_Taxonomy <- NULL
field_merged$field_taxo <- NULL
```

```
colSums(field_merged)
```

```
##           field_commdesc           field_map
##           147             46
##           field_impass           field_troph
##           67             37
##           field_dist           field_physio
##           170             124
##           field_phylo           field_Molecular_ecology
##           31             121
## field_environmental_drivers
##           137
```

```
field_merged <- field_merged[, order(colSums(field_merged), decreasing = F)]
```

```
colSums(field_merged)
```

```
##                field_phylo                field_troph
##                31                37
##                field_map                field_impass
##                46                67
##    field_Molecular_ecology                field_physio
##                121                124
## field_environmental_drivers                field_commdesc
##                137                147
##                field_dist
##                170
```

Rename fields with long version names

```
field_merged %<>%
  rename(`Biodiversity and \n community structure (146)` = field_commdesc,
         `Distribution and \n connectivity (171)` = field_dist,
         `Disturbances (66)` = field_impass,
         `Mapping (46)` = field_map,
         `Physiology (124)` = field_physio,
         `Phylogenesis (31)` = field_phylo,
         `Trophic ecology (37)` = field_troph,
         `Molecular ecology (120)` = field_Molecular_ecology,
         `Environmental drivers and \n spatio-temporal patterns (136)` = field_environmental_drivers
  )
```

```
colSums(field_merged)
```

```
##                Phylogenesis (31)
##                31
##                Trophic ecology (37)
##                37
##                Mapping (46)
##                46
##                Disturbances (66)
##                67
##                Molecular ecology (120)
##                121
##                Physiology (124)
##                124
## Environmental drivers and \n spatio-temporal patterns (136)
##                137
##                Biodiversity and \n community structure (146)
##                147
##                Distribution and \n connectivity (171)
##                170
```

Prepare the indicator matrix


```
ind_matrix <- split_results[["indicators"]]
```

Select only the most common indicators

```
colSums(ind_matrix)
```

```
##      Abundance      Cover      Diversity      Habitat
##      93          135          180          57
## Morphological complexity      Size geneticSequences
##      143          3          70          120
##      Density      taxo      Biomass      Trophic
##      36          90          26          37
##      rugosity      Mortality      Physiological      growth
##      1          15          94          22
##      Recruitment      surv      sex      Health
##      14          16          7          28
##      Behaviour      NIS      disease      impact
##      17          8          6          5
##      ALDFGs      Litter      func      fecundity
##      3          4          10          1
```

```
ind_common_matrix <- ind_matrix[, colSums(ind_matrix) > 30]
```

```
colSums(ind_common_matrix)
```

```
##      Abundance      Cover      Diversity      Habitat
##      93          135          180          57
## Morphological      Size geneticSequences      Density
##      143          70          120          36
##      taxo      Trophic      Physiological
##      90          37          94
```

```
ind_common_matrix <- ind_common_matrix[, order(colSums(ind_common_matrix), decreasing = TRUE)]
```

```
ind_common_matrix <- as.data.frame(ind_common_matrix)
```

```
ind_common_matrix <- ind_common_matrix[!purrr::map_lgl(ind_common_matrix, ~ all(is.na(.)))] # remove empty column
```

Remove taxonomy studies which have a dedicated section

```
ind_common_matrix$taxo <- NULL
```

```
colSums(ind_common_matrix)
```

```
##      Diversity      Morphological      Cover geneticSequences
##      180          143          135          120
##      Physiological      Abundance      Size      Habitat
##      94          93          70          57
```

```
##          Trophic          Density
##          37             36
```

Rename indicators with long version names

```
ind_common_matrix %<>%
  rename(`Abundance \n (92)` = Abundance,
         `Cover \n (134)` = Cover,
         `Diversity \n (179)` = Diversity,
         `Habitat \n (56)` = Habitat,
         `Morphology \n (142)` = Morphological,
         `Size \n (70)` = Size,
         `Density \n (36)` = Density,
         `Physiological \n indicator \n (94)` = Physiological,
         `Trophic \n (37)` = Trophic,
         `Genetic \n sequence \n (120)` = geneticSequences
  )
```

```
colSums(ind_common_matrix)
```

```
##          Diversity \n (179)          Morphology \n (142)
##                      180                      143
##          Cover \n (134)          Genetic \n sequence \n (120)
##                      135                      120
## Physiological \n indicator \n (94)          Abundance \n (92)
##                      94                      93
##          Size \n (70)          Habitat \n (56)
##                      70                      57
##          Trophic \n (37)          Density \n (36)
##                      37                      36
```

Multiply the transpose of the field matrix with the indicator matrix

```
co_occurrence_matrix <- t(as.matrix(field_merged)) %*% as.matrix(ind_common_m
atrix)
```

Reshape the co-occurrence matrix for plotting}

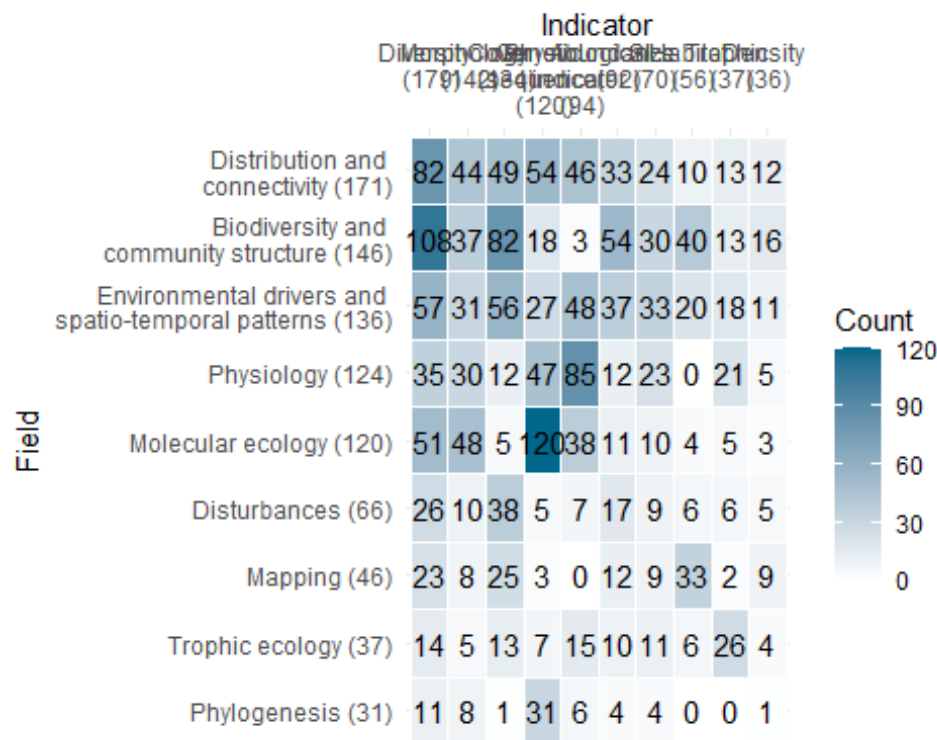
```
co_occurrence_df <- reshape2::melt(co_occurrence_matrix)

colnames(co_occurrence_df) <- c("Field", "Indicator", "Count")
```

Plot the heatmap

```
field_ind_heat <- ggplot(co_occurrence_df, aes(x = Indicator, y = Field, fill
= Count)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "white", high = "deepskyblue4") +
  geom_text(aes(label = Count), color = "black") +
  theme_minimal()+
  scale_x_discrete(position="top")
```

field_ind_heat



Export plot

```
ggsave("field_ind_heat.png",
       units = "in", width = 12.5, height = 5.5,
       path = here ("Extracted_data_analysis", "tropical",
                    "Output-Visualization"))
```

Calculate # of ecological indicators per record

Create a column with the sum of ecological indicators values to see how many different sampling units were adopted by each study

```
ind_summary <- as.data.frame(split_results[["indicators"]])

ind_summary <- ind_summary[!purrr::map_lgl(ind_summary, ~ all(is.na(.)))] # remove empty column

ind_summary %<>% rowwise() %>%
  mutate(ind_per_record = sum(c_across(where(is.numeric))))
```

Calculate number of records by number of sampling units in %

```
ind_summary$ind_per_record <- as.factor(ind_summary$ind_per_record)
```

```
ind_per_record <- ind_summary %>%
  group_by(ind_per_record) %>%
  summarise(n_records = length(ind_per_record)) %>%
  mutate(percentage= round(n_records / sum(n_records) * 100,1))
```

```
ind_per_record
```

```
## # A tibble: 10 × 3
##   ind_per_record n_records percentage
##   <fct>          <int>         <dbl>
## 1 0              3          0.7
## 2 1             67         15.1
## 3 2            135         30.3
## 4 3            140         31.5
## 5 4             49          11
## 6 5             31           7
## 7 6             11          2.5
## 8 7              6          1.3
## 9 8              2          0.4
## 10 9             1          0.2
```

SAMPLING UNIT

Prepare a data frame for sampling unit

```
sampling_df <- tropical %>%
  filter(is.na(taxonomic_publication)) %>% # remove purely taxonomic studies
  select(File,recordTitle,DOI,publicationYear,mininumSurveyedDepthInMeters,maximumSurveyedDepthInMeters,samplingUnit)
```

```
sampling_df %>% tidyr::drop_na(samplingUnit)
sampling_df[!is.na(sampling_df$samplingUnit),]
```

```
sampling_df %<>%
  filter(samplingUnit != "notapp")
```

```
sampling_df$samplingUnit <- as.character(sampling_df$samplingUnit)
```

Prepare columns per sampling units

```
sampling_lvls <- unique(unlist(str_split(sampling_df$samplingUnit,
                                          "\\+"))
                        )

rownames(is.na(sampling_df$samplingUnit))

## NULL
```

```
sampling_df[,sampling_lvls] <- sapply(sampling_lvls,
                                     function(y){
                                       str_count(sampling_df$samplingUnit, y)
                                     })

rm(sampling_lvls)

colnames(sampling_df)

## [1] "File" "recordTitle"
## [3] "DOI" "publicationYear"
## [5] "mininumSurveyedDepthInMeters" "maximumSurveyedDepthInMeters"
## [7] "samplingUnit" "quadrat"
## [9] "sample" "transect"
## [11] "permquad" "pit"
## [13] "permtrans" "expunit"
## [15] "lit" "nd"
## [17] "permlit" "roaming_transect"
## [19] "random_swim" "arms"
## [21] "plotless" "stationary"
## [23] "radial_transect"
```

Calculate # of sampling unit per record

Create a column with the sum of sampling units values to see how many different sampling units were adopted by each study

```
sampling_df %<>%
  mutate(type_per_record = rowSums(across(quadrat:permlit))
)
)
```

Calculate number of records by number of sampling units in %

```
sampling_df$type_per_record <- as.factor(sampling_df$type_per_record)

sampling_unit_per_record <- sampling_df %>%
  group_by(type_per_record) %>%
  summarise(n_records = length(type_per_record)) %>%
  mutate(percentage= round(n_records / sum(n_records) * 100,0))

sampling_unit_per_record

## # A tibble: 4 × 3
##   type_per_record n_records percentage
##   <fct>          <int>      <dbl>
## 1 0              5          1
## 2 1             285         76
```

## 3 2	71	19
## 4 3	15	4

Create a data frame with number of publication per sampling unit type

```
sampling_unit_tot <- sampling_df %>%
  summarise(across(c(8:(length(sampling_df)-1)),
    list(sum)))

sampling_unit_tot <- reshape2::melt(sampling_unit_tot)

## No id variables; using all as measure variables

colnames(sampling_unit_tot) <- c("Sampling_unit", "Count")

sampling_unit_tot %<>%
  mutate_at("Sampling_unit",
    stringr::str_replace, "_1", "")
```

Simplified sampling units

Create a column with a simplified sampling units for transect and quadrat

```
sampling_unit_tot %<>%
  mutate(Sampling_unit_macro = case_when(Sampling_unit == 'quadrat' |
    Sampling_unit == "permquad" ~ "Quadrat",
    Sampling_unit == "transect" |
    Sampling_unit == "lit" |
    Sampling_unit == "pit" |
    Sampling_unit == "permlit" |
    Sampling_unit == "permtrans" |
    Sampling_unit == "circular_transect" |
    Sampling_unit == "radial_transect" ~ "Transect",

    Sampling_unit == "nd" ~ "ND",
    Sampling_unit == "stationary" ~ "Stationary",

    Sampling_unit == "expunit" ~ "Experimental",

    Sampling_unit == "sample" ~ "Sample",
    .default = "Other")

# Change columns order
sampling_unit_tot <- cbind(sampling_unit_tot$Sampling_unit_macro,
  sampling_unit_tot$Sampling_unit,
  sampling_unit_tot$Count)

colnames(sampling_unit_tot) <- c("Sampling_unit_macro", "Sampling_unit", "Count")
```

```
sampling_unit_tot <- as.data.frame(sampling_unit_tot)

sampling_unit_tot[,1:2] <- lapply(sampling_unit_tot[,1:2], as.factor)
sampling_unit_tot$Count <- as.numeric(sampling_unit_tot$Count)

sampling_unit_simplified <- aggregate(Count ~ Sampling_unit_macro, data = sampling_unit_tot, FUN = sum)

sampling_unit_simplified %<>%
  group_by(Sampling_unit_macro) %>%
  mutate(Percentage_macro= round(Count / sum(sampling_unit_simplified$Count)
* 100,1))

sampling_unit_simplified

## # A tibble: 7 × 3
## # Groups:   Sampling_unit_macro [7]
##   Sampling_unit_macro Count Percentage_macro
##   <fct>           <dbl>           <dbl>
## 1 Experimental unit     46             9.3
## 2 ND                   21             4.3
## 3 Other                 10              2
## 4 Quadrat              111            22.5
## 5 Sample               166            33.7
## 6 Stationary            10              2
## 7 Transect             129            26.2
```

Build a treemap with simplified sampling unit

```
library(treemapify)

RdYlBu <- RColorBrewer::brewer.pal(7, "RdYlBu")
RdYlBu

## [1] "#D73027" "#FC8D59" "#FEE090" "#FFFFBF" "#E0F3F8" "#91BFDB" "#4575B4"

sampling_simplified_treemap <-
ggplot(sampling_unit_simplified,
  aes(area=Count,
    fill=Sampling_unit_macro,
    line = "black",
    label=paste0(Sampling_unit_macro, " \n ",Percentage_macro,"%"))
)+
treemapify::geom_treemap(layout = "squarified")+
geom_treemap_text(place = "centre",
  size = 20)+
theme(plot.title = element_text(hjust = 0.5, size= 20),
  legend.position = "none")+
scale_fill_manual(values= c("#FC8D59", "#FEE090", "#FFFFBF", "#E0F3F8", "#D73027", "#91BFDB", "#4575B4"))
```

```

3027", "#91BFD8", "#4575B4"))
  scale_fill_brewer(palette = "RdYlBu")

## <ggproto object: Class ScaleDiscrete, Scale, gg>
##   aesthetics: fill
##   axis_order: function
##   break_info: function
##   break_positions: function
##   breaks: waiver
##   call: call
##   clone: function
##   dimension: function
##   drop: TRUE
##   expand: waiver
##   fallback_palette: function
##   get_breaks: function
##   get_breaks_minor: function
##   get_labels: function
##   get_limits: function
##   get_transformation: function
##   guide: legend
##   is_discrete: function
##   is_empty: function
##   labels: waiver
##   limits: NULL
##   make_sec_title: function
##   make_title: function
##   map: function
##   map_df: function
##   minor_breaks: waiver
##   n.breaks.cache: NULL
##   na.translate: TRUE
##   na.value: NA
##   name: waiver
##   palette: function
##   palette.cache: NULL
##   position: left
##   range: environment
##   rescale: function
##   reset: function
##   train: function
##   train_df: function
##   transform: function
##   transform_df: function
##   super: <ggproto object: Class ScaleDiscrete, Scale, gg>

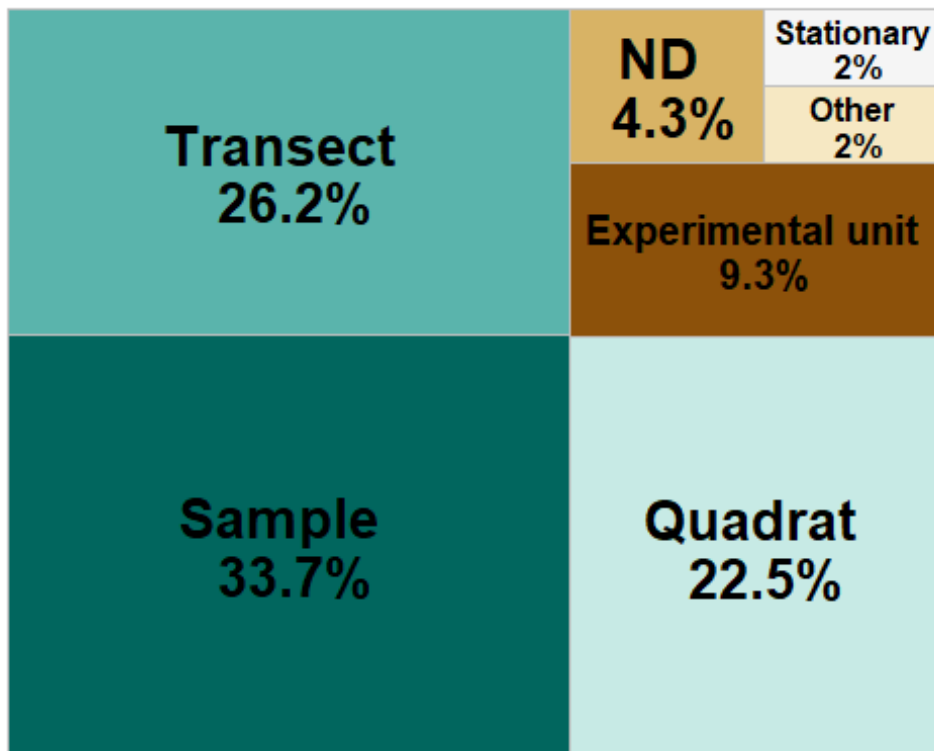
sampling_unit_simplified %<>%
  mutate(case_when(Sampling_unit_macro == "Experimental unit" ~
    "Experimental \n unit", T ~ Sampling_unit_macro))

```



```
sampling_simplified_treemap <-
ggplot(sampling_unit_simplified,
  aes(area=Count,
    fill= reorder(Sampling_unit_macro, -Count),
    line = "black",
    label=paste0(Sampling_unit_macro, " \n ", Percentage_macro, "%"))
  )+
treemapify::geom_treemap(layout = "squarified")+
geom_treemap_text(place = "centre", fontface = "bold",
  size = 20)+
  theme(plot.title = element_text(hjust = 0.5, size= 20),
    legend.position = "none")+
  scale_fill_manual(values=c("#01665E", "#5AB4AC", "#C7EAE5", "#8C510A", "#D8B365", "#F6E8C3", "#F5F5F5")) # selected from "BrBG" palette of RBrewer

sampling_simplified_treemap
```



Export sampling_simplified_treemap

```
ggsave("sampling_simplified_treemap.png",
  plot = sampling_simplified_treemap,
  units = "in", width = 7, height = 6,
  path = here ("Extracted_data_analysis",
    "tropical",
    "Output-Visualization"))
```

Export sampling_unit_tot table for supplementary materials

```
library(openxlsx)

openxlsx::write.xlsx(sampling_unit_tot,
                     file = here("Extracted_data_analysis",
                                "Output-Transformed_data",
                                "sampling_unit_tot.xlsx"))
```

Quadrat extracted from transect

```
print(paste("Number of publications where quadrat is the sampling unit or one
of the sampling unit: ",
            nrow(quad_design <- subset(tropical, grepl("quad", samplingUnit)
                                     )
            )
      )

## [1] "Number of publications where quadrat is the sampling unit or one of t
he sampling unit: 110"

print(paste("Number of publications where quadrats were collected along trans
ects: ", nrow(subset(quad_design,
                     grepl("transect\\(quadrat", meso_design) |
                     grepl("transect\\(depth\\(quadrat", meso_design)
                     )
                     )
      )

## [1] "Number of publications where quadrats were collected along transects:
81"
```

DATA TYPE

```
data_type <- count_results[["collectedDataType"]]

data_type$Percentage <- round(data_type$Percentage, 1)

data_type

## # A tibble: 29 × 3
##   collectedDataType      Count Percentage
##   <fct>              <int>      <dbl>
## 1 photo              49         11
## 2 photo+specimen     51        11.5
## 3 photo+specimen+sediment 1         0.2
## 4 photo+specimen+sediment+video 3         0.7
## 5 photo+specimen+video 9          2
## 6 photo+specimen+video+visual 4         0.9
## 7 photo+specimen+visual 7         1.6
```

```
## 8 photo+specimen+water      1      0.2
## 9 photo+video              14      3.1
## 10 photo+video+visual      4      0.9
## # i 19 more rows
```

Lump uncommon combination into “Other”

```
data_type_simplified <- data_type %>%
  rename(`Data type` = collectedDataType) %>%
  mutate(`Data type` = as.character(`Data type`)) %>%
  filter(Percentage > 2) %>%
  bind_rows(
    data_type %>%
      filter(Percentage <= 2) %>%
      summarise(
        `Data type` = "other",
        Count = sum(.data$Count),
        Percentage = sum(.data$Percentage)
      )
  )

data_type_simplified %<>%
  mutate(`Data type` =
    case_when(`Data type` == 'photo' ~ "Photo",
              `Data type` == "video" ~ "Video",
              `Data type` == "other" ~ "Other",
              `Data type` == "specimen" ~ "Specimen",
              `Data type` == "photo+specimen" ~ "Photo + \n Specimen",
              `Data type` == "photo+specimen+video" ~ "Photo + Video + \n Specimen",
              `Data type` == "photo+video" ~ "Photo + \n Video",
              `Data type` == "photo+visual" ~ "Photo + \n Visual",
              `Data type` == "specimen+visual" ~ "Specimen + \n Visual",
              `Data type` == "specimen+video" ~ "Specimen + \n Video",
              .default = "Other")
  )
```

Build a treemap with data type

```
library(treemapify)
library(RColorBrewer)

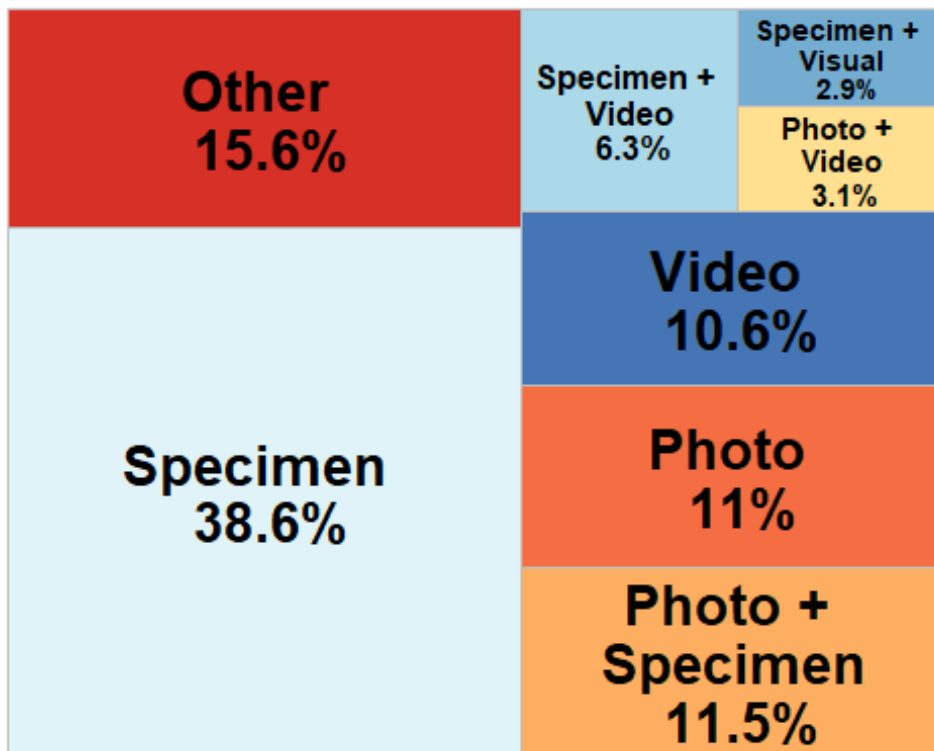
data_type_treemap <-
  ggplot(data_type_simplified,
    aes(area=Count,
        fill=`Data type`,
```

```

    line = "black",
    label=paste0(`Data type`, " \n ",Percentage,"%"))
  )+
  treemapify::geom_treemap(layout = "squarified")+
  geom_treemap_text(place = "centre", fontface = "bold",
    size = 20)+
  theme(plot.title = element_text(hjust = 0.5, size= 20),
    legend.position = "none")+
  scale_fill_brewer(palette = "RdYlBu")

```

data_type_treemap



Export sampling_simplified_treemap

```

ggsave("data_type_treemap.png",
  plot = data_type_treemap,
  units = "in", width = 7, height = 6,
  path = here ("Extracted_data_analysis",
    "Output-Visualization"))

```

"tropic"

HABIT

```
habit <- as.data.frame(count_results["habit"])
```

```

# Rename columns
habit %<>%

```

```

rename(Habit = habit.habit,
       Count = habit.Count,
       Percentage = habit.Percentage)

```

Make the habit donut plot

```

habit %<>%
  mutate(ymax = cumsum(Count),
         ymin = c(0, head(ymax, n = -1)),
         mid = (ymin + ymax) / 2,
         angle = 90 - (mid / sum(Count)) * 360
  )

# Split and assign labels
habit_inside <- habit %>%
  mutate(label = paste0(round(Percentage,1), "%"))

habit_outside <- habit %>%
  mutate(label = Habit)

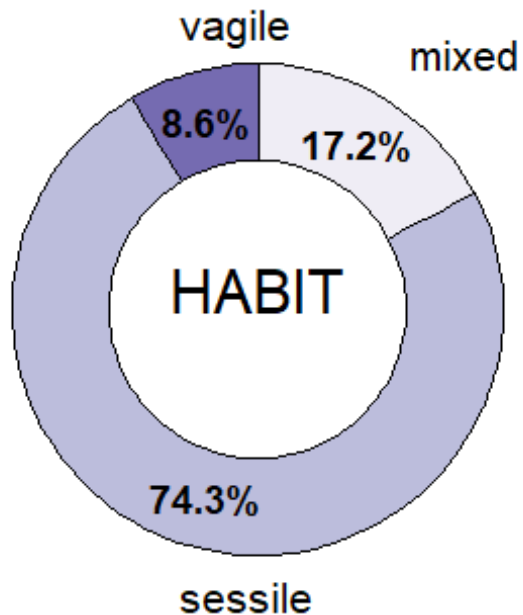
# Plot
habit_donut <- ggplot() +
  # Donut segments
  geom_rect(data = habit,
           aes(ymin = ymin, ymax = ymax, xmin = 2.5, xmax = 4, fill = Habit),
           color = "black") +

  # Inside labels
  geom_text(data = habit_inside,
           aes(x = 3.2, y = mid, label = label),
           size = 5, color = "black",
           fontface = "bold") +

  # Outside labels with Habit name
  ggrepel::geom_text_repel(data = habit_outside,
                          aes(x = 4.3, y = mid, label = label),
                          size = 5.5,
                          nudge_x = 0.5,
                          direction = "y",
                          segment.color = "black",
                          hjust = 0,
                          box.padding = 0.3,
                          max.overlaps = Inf) +
  coord_polar(theta = "y") +
  xlim(0.2, 5) +
  scale_fill_brewer(palette = "Purples") +
  theme_void() +
  theme(legend.position = "none")+
  annotate(geom = 'text', x = 0.5, y = 0,
         label = c("HABIT"), size = 8)

```

habit_donut



Export habit_donut

```
ggsave("habit_donut.png",  
  plot = habit_donut,  
  units = "in", width = 7, height = 6,  
  path = here ("Extracted_data_analysis",  
    "Output-Visualization"))
```

SUBSTRATE

```
substrate <- as.data.frame(count_results["meso_substrate"])  
  
# Rename columns  
substrate %<>%  
  rename(Substrate = meso_substrate.meso_substrate,  
    Count = meso_substrate.Count,  
    Percentage = meso_substrate.Percentage)  
  
# Remove rows with nd  
substrate <- substrate[!substrate$Substrate %in% "nd",]  
  
# Merge substrate combination with less than 3% and recalculate Count and Percentages
```

```

substrate <- substrate %>%
  mutate(Substrate = if_else(Percentage < 3, "Other combinations", Substrate))
%>%
  group_by(Substrate) %>%
  select(-Percentage) %>%
  summarise(
    Count = sum(Count)) %>%
  mutate(Percentage = round(Count / sum(Count) * 100,1)
  )

substrate %<>%
  mutate(
    Substrate = case_when(Substrate == "hard" ~ "Hard",
                          Substrate == "soft" ~ "Soft",
                          Substrate == "mixed" ~ paste0("Hard+", "\n", "Soft"),
                          Substrate == "artificial" ~ "Artificial",
                          Substrate == "Other combinations" ~ "Other combinat
ions"))

```

Make the substrate donut plot

```

substrate %<>%
  mutate(
    ymax = cumsum(Count),
    ymin = c(0, head(ymax, n = -1)),
    mid = (ymin + ymax) / 2,
    angle = 90 - (mid / sum(Count)) * 360
  )

# Split and assign labels
substrate_inside <- substrate %>%
  filter(Percentage >= 5) %>%
  mutate(label = paste0(Percentage, "%"))

substrate_outside <- substrate %>%
  mutate(label = Substrate)

# Plot
substrate_donut <- ggplot() +
  # Donut segments
  geom_rect(data = substrate,
            aes(ymin = ymin, ymax = ymax, xmin = 2.5, xmax = 4, fill = Substrate),
            color = "black") +

  # Inside labels
  geom_text(data = substrate_inside,
            aes(x = 3.2, y = mid, label = label),

```

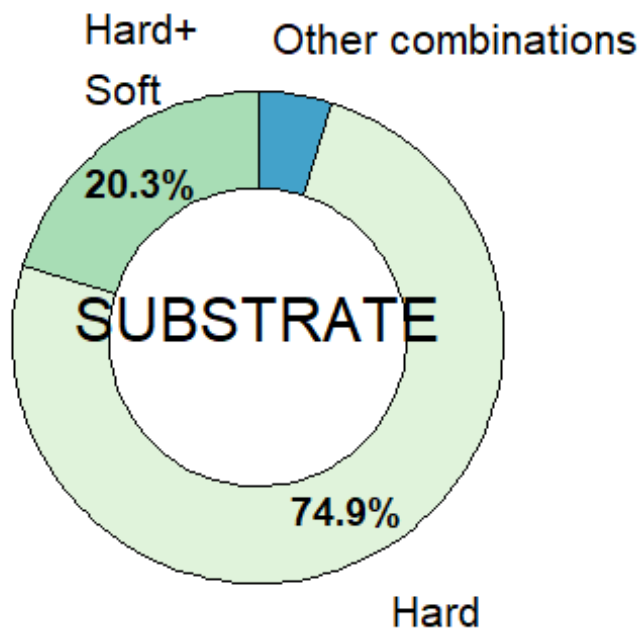
```

    size = 5, color = "black",
    fontface = "bold") +

# Outside labels with substrate name
ggrepel::geom_text_repel(data = substrate_outside,
    aes(x = 4.4, y = mid, label = label),
    size = 5.5,
    nudge_x = 0.3,
    direction = "y",
    segment.color = "black",
    hjust = 0,
    box.padding = 0.3,
    max.overlaps = Inf) +
coord_polar(theta = "y") +
xlim(0.2, 5) +
scale_fill_brewer(palette = "GnBu") +
theme_void() +
theme(legend.position = "none")+
annotate(geom = 'text', x = 0.5, y = 0,
    label = c("SUBSTRATE"), size = 8)

```

substrate_donut



Export the

substrate donut plot

```

ggsave("substrate_donut.png",
    plot = substrate_donut,

```



```

units = "in", width = 7, height = 6,
path = here ("Extracted_data_analysis",
al",
"Output-Visualization"))
"tropic

```

BENTHIC POSITION

Create a data frame with benthic position

```

position <- as.data.frame(count_results["environmentalPosition"])

# Rename columns
position %<>%
  rename(Position = environmentalPosition.environmentalPosition,
          Count = environmentalPosition.Count,
          Percentage = environmentalPosition.Percentage)

# Merge position combination with less than 3% and recalculate Count and Percentages
position <- position %>%
  mutate(Position = if_else(Percentage < 1, "Other combinations", Position))
  %>%
  group_by(Position) %>%
  select(-Percentage) %>%
  summarise(
    Count = sum(Count)) %>%
  mutate(Percentage = round(Count / sum(Count) * 100,1)
  )

```

Make the benthic position donut plot

```

position %<>%
  mutate(
    Position = case_when(Position == "epi" ~ "Epi",
                        Position == "endo" ~ "Endo",
                        Position == "dem+epi" ~ "Hyper + \n Epi",
                        Position == "endo+epi" ~ "Endo + \n Epi",
                        Position == "Other combinations" ~ "Other combinations"),
    ymax = cumsum(Count),
    ymin = c(0, head(ymax, n = -1)),
    mid = (ymin + ymax) / 2,
    angle = 90 - (mid / sum(Count)) * 360
  )

# Split and assign labels
position_inside <- position %>%
  filter(Percentage > 3) %>%
  mutate(label = paste0(Percentage, "%"))

```

```

position_outside <- position %>%
  mutate(label = Position)

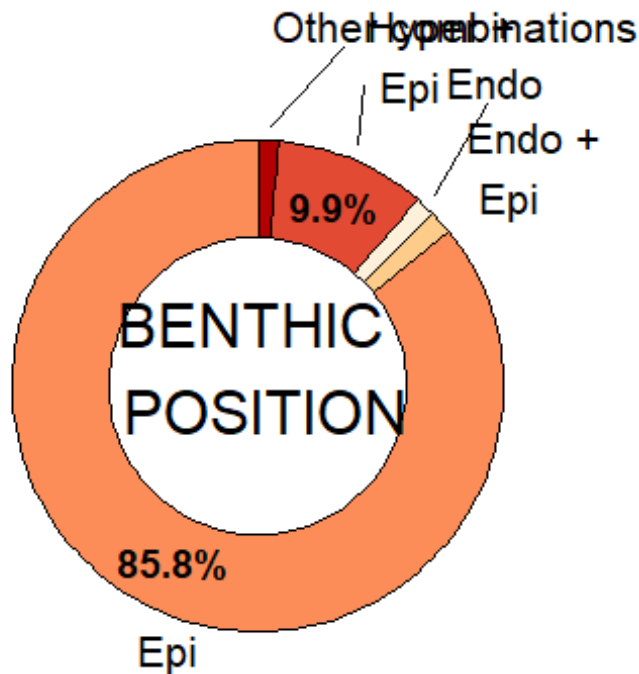
# Plot
benthic_position_donut <- ggplot() +
  # Donut segments
  geom_rect(data = position,
            aes(ymin = ymin, ymax = ymax, xmin = 2.5, xmax = 4, fill = Position),
            color = "black") +

  # Inside labels
  geom_text(data = position_inside,
            aes(x = 3.2, y = mid, label = label),
            size = 5, color = "black",
            fontface = "bold") +

  # Outside labels with Position name
  ggrepel::geom_text_repel(data = position_outside,
                           aes(x = 4.2, y = mid, label = label),
                           size = 5.5,
                           nudge_x = 0.3,
                           direction = "y",
                           segment.color = "black",
                           hjust = 0,
                           box.padding = 0.3,
                           max.overlaps = Inf) +
  coord_polar(theta = "y") +
  xlim(0.2, 5) +
  scale_fill_brewer(palette = "OrRd") +
  theme_void() +
  theme(legend.position = "none")+
  annotate(geom = 'text', x = 0.5, y = 0,
           label = c("BENTHIC \n POSITION"), size = 8)

benthic_position_donut

```



Export the benthic_position donut plot

```
ggsave("benthic_position_donut.png",
       plot = benthic_position_donut,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",
                    "Output-Visualization"))
```

TARGET

Target level

```
target_level <- as.data.frame(count_results["taxonomicTargetLevel"])

# Rename columns
target_level %<>%
  rename(Level = taxonomicTargetLevel.taxonomicTargetLevel,
         Count = taxonomicTargetLevel.Count,
         Percentage = taxonomicTargetLevel.Percentage)

# Remove rows with nd
target_level <- target_level[!target_level$Level %in% "species,species",]
target_level <- target_level[!target_level$Level %in% ",species",]

# Merge habit combination with less than 3% and recalculate Count and Percent
ages
```

```

target_level <- target_level %>%
  mutate(Level = if_else(Percentage < 3, "Other combinations", Level)) %>%
  group_by(Level) %>%
  select(-Percentage) %>%
  summarise(
    Count = sum(Count)) %>%
  mutate(Percentage = round(Count / sum(Count) * 100,1)
  )

```

Treemap with the taxonomic target level

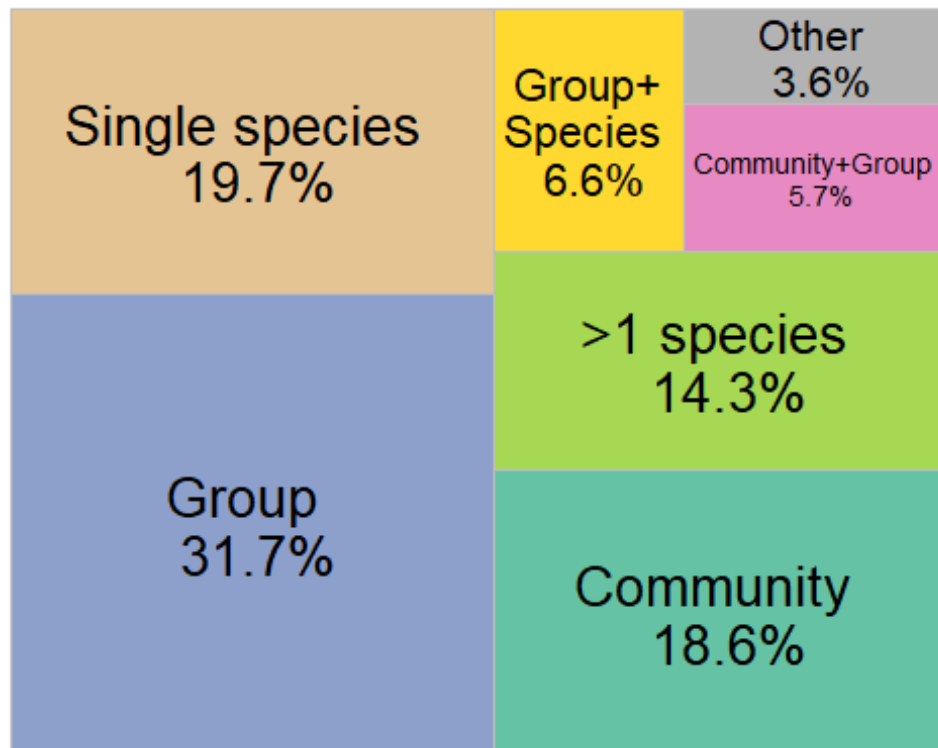
```

target_level_labels <- target_level %<>%
  mutate(
    Level = case_when(Level == "comm" ~ "Community",
                      Level == "group" ~ "Group",
                      Level == "single" ~ "Single species",
                      Level == "species" ~ ">1 species",
                      Level == "comm+group" ~ "Community+Group",
                      Level == "comm+species" ~ "Community+Species",
                      Level == "group+species" ~ "Group+\nSpecies",
                      Level == "Other combinations" ~ "Other")
  )

target_level_treemap <- ggplot(target_level_labels,
  aes(area=Count,
    fill=Level,
    line = "black",
    label=paste0(Level,
      "\n ", Percentage, "%"))
  )+
  treemapify::geom_treemap(layout = "squarified")+
  treemapify::geom_treemap_text(place = "centre",
    size = 20)+
  theme(plot.title = element_text(hjust = 0.5, size= 20),
    legend.position = "none")+
  scale_fill_manual(values= c("#A6D854" , "#66C2A5", "#E78AC3", "#8DA0CB", "#FFD92F", "#B3B3B3", "#E5C494", "#FC8D62"))

target_level_treemap

```



```
ggsave("target_level_treemap.png",
  plot = target_level_treemap,
  units = "in", width = 9, height = 6,
  path = here ("Extracted_data_analysis",
    "Output-Visualization"))
```

"tropic"

Detected phyla in community studies

```
detected <- tropical %>%
  filter(str_detect(taxonomicTargetLevel, "comm")) %>%
  select(File,
    detected_meso_cnidaria,
    detected_meso_sponges,
    detected_meso_seagrass,
    detected_meso_algae,
    detected_meso_mollusca,
    detected_meso_chondrichthyes,
    detected_meso_fish,
    detected_meso_crustacea,
    detected_meso_echino,
    detected_meso_other_taxa)

detected_lvls <-
  unique(
    unlist(
      str_split(detected$detected_meso_other_taxa, "\\+")
    )
  )
```

```

    )
detected_lvls<- detected_lvls[! detected_lvls %in% c("",NA,"notapp","", "nd")]
detected[,detected_lvls] <- sapply(detected_lvls,
                                   function(y){
                                     str_count(detected$detected_meso_oth
r_taxa, y)
                                   }
                                   )

rm(detected_lvls)

detected %<>%
  mutate(across(everything(), as.character)) %>%
  replace(is.na(.), "0") %>%
  select(-detected_meso_other_taxa) %>%
  filter(if_all(everything(), ~ !grepl("notapp", .x)))

detected <- detected %>%
  mutate(across(-File, ~ case_when(
    .x %in% c("x", "x", "xx") ~ "1",
    .x == "", ~ "0",
    TRUE ~ .x
  ))) %>%
  mutate(across(-File, as.numeric))

```

Calculate number of publications per taxonomic group

```

detected_tot <-
  detected %>%
  summarise(across(where(is.numeric),
                    list(sum)
                  )
            )

detected_tot <- reshape2::melt(detected_tot, id.vars = NULL)

colnames(detected_tot) <- c("Taxa", "Count")

detected_tot %<>%
  mutate_at("Taxa",
            stringr::str_replace, "_1", "") %>%
  mutate_at("Taxa",
            stringr::str_replace, "detected_meso_", "")

detected_tot %<>%
  mutate(Percentage= round(Count / nrow(detected) * 100,1))

rm(detected)

```

Merge poorly represented taxonomic groups into “Other”

```
detected_tot_collapsed <- detected_tot %>%  
  mutate(Taxa = if_else(Percentage < 3, "Other", Taxa)) %>%  
  group_by(Taxa) %>%  
  summarise(  
    Count = sum(Count),  
    Percentage = sum(Percentage)  
  )
```

Create a bar chart with detected taxonomic groups

```
detected_bar <- ggplot(detected_tot_collapsed, aes(x = reorder(Taxa, Count),  
y = Count, fill= Taxa)) +  
  geom_bar(stat = "identity") +  
  geom_text(  
    data = detected_tot_collapsed %>% filter(Percentage > 11),  
    aes(label = paste0(round(Percentage, 0), "%")),  
    position = position_stack(vjust = 0.5),  
    color = "black",  
    size = 3.5  
  ) +  
  labs(y = "# Publications at the community level", x = "Detected taxonomic group") +  
  theme_minimal() +  
  coord_flip() + # optional: flips for horizontal bars  
  theme(legend.position = "none") +  
  scale_x_discrete(labels = rev(c("Cnidaria",  
    "Porifera",  
    "macroalgae*",  
    "Osteichthyes",  
    "Echinodermata",  
    "Other",  
    "Mollusca",  
    "Crustacea",  
    "Tunicata",  
    "Bryozoa",  
    "Chondrichthyes",  
    "Annelida",  
    "Cyanobacteria"))))  
  
detected_palette <- c("algae" = "darkseagreen3",  
  "cnidaria" = "mediumvioletred",  
  "crustacea" = "indianred1",  
  "sponges" = "khaki1",  
  "mollusca" = "darkorange3",  
  "fish" = "paleturquoise2",  
  "echino" = "royalblue4",  
  "Tunicata" = "mediumpurple4",  
  "chondrichthyes" = "gray69",
```

```

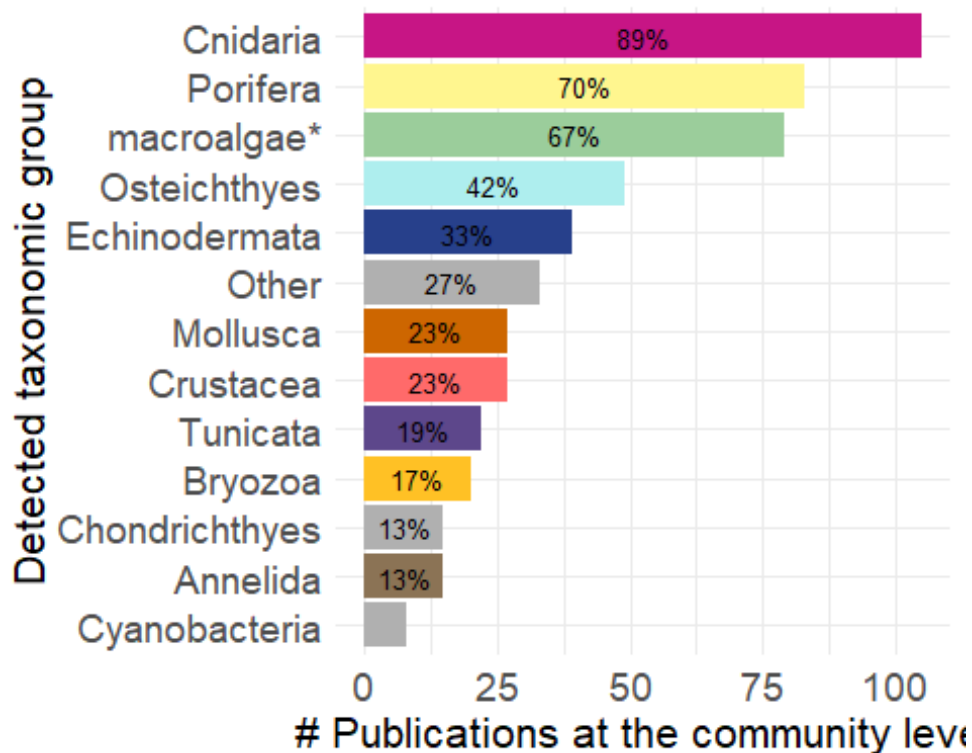
"Brachiopoda" = "gray69",
"Other" = "gray69",
"Cyanobacteria" = "gray69",
"Bryozoa" = "goldenrod1",
"Annelida" = "burlywood4")

```

```

detected_bar +
  scale_fill_manual(values = detected_palette,
    limits = names(detected_palette))+
  theme(axis.text = element_text(size = 14),
    axis.title = element_text(size = 16)
  )

```



```

ggsave("detected_bar.png",
  units = "in", width = 7, height = 6,
  path = here ("Extracted_data_analysis",
    "Output-Visualization"))

```

Target taxonomic groups not at the community level

```

group <- tropical %>%
  filter(taxonomicTargetLevel != "comm") %>%
  select(File,
    detected_meso_cnidaria,
    detected_meso_sponges,
    detected_meso_seagrass,

```



```

    detected_meso_algae,
    detected_meso_mollusca,
    detected_meso_chondrichthyes,
    detected_meso_fish,
    detected_meso_crustacea,
    detected_meso_echino,
    detected_meso_other_taxa,
    targetTaxa)

group_lvls <-
  unique(
    unlist(
      str_split(group$detected_meso_other_taxa, "\\+")
    )
  )
group_lvls<- group_lvls[! group_lvls %in% c("",NA,"notapp",",","nd")]
group[,group_lvls] <- sapply(group_lvls,
                             function(y){
                               str_count(group$detected_meso_other_t
axa, y)
                             })

rm(group_lvls)

group <- group %>%
  mutate(across(everything(), as.character)) %>%
  replace(is.na(.), "0") %>%
  select(-detected_meso_other_taxa,-targetTaxa) %>%
  filter(if_all(everything(), ~ !grepl("notapp", .x))) %>%
  mutate(across(-File, ~ case_when(
    .x %in% c("x", "x", "xx") ~ "1",
    .x == ", " ~ "0",
    TRUE ~ .x
  ))) %>%
  mutate(across(-File, as.numeric))

group_tot <-
  group %>%
  summarise(across(where(is.numeric),
                    list(sum)
                  )
            )

group_tot <- reshape2::melt(group_tot, id.vars = NULL)

colnames(group_tot) <- c("Taxa", "Count")

group_tot %<>%
  mutate_at("Taxa",

```

```

      stringr::str_replace, "_1", "") %>%
mutate_at("Taxa",
      stringr::str_replace, "detected_meso_", "")

group_tot %<>%
  mutate(Percentage= round(Count / nrow(group) * 100,1))

group_tot_collapsed <- group_tot %>%
  mutate(Taxa = if_else(Percentage < 3, "Other", Taxa)) %>%
  group_by(Taxa) %>%
  summarise(
    Count = sum(Count),
    Percentage = sum(Percentage),
    .groups = "drop"
  )

group_tot_collapsed %<>%
  mutate(Taxa = case_when(Taxa == 'echino' ~ "Echinodermata",
                          Taxa == 'fish' ~ "Osteichthyes",
                          Taxa == "Symbiodiniaceae" ~ "Symbiodiniaceae",
                          Taxa == "sponges" ~ "Porifera",
                          Taxa == "crustacea" ~ "Crustacea",
                          Taxa == "cnidaria" ~ "Cnidaria",
                          Taxa == "algae" ~ "macroalgae*",
                          .default = Taxa))

group_bar <- ggplot(group_tot_collapsed, aes(x = reorder(Taxa, Count), y = Count, fill= Taxa)) +
  geom_bar(stat = "identity") +
  geom_text(
    data = group_tot_collapsed %>% filter(Percentage > 10),
    aes(label = paste0(round(Percentage, 0), "%"),
        position = position_stack(vjust = 0.5),
        color = "black",
        size = 3.5
    )+
  labs(y = "N of publications not at community level", x = "Taxonomic group")
+
  theme_minimal() +
  coord_flip()+ # optional: flips for horizontal bars
  theme(legend.position = "none")

group_palette <- c("macroalgae*" = "darkseagreen3",
                  "Cnidaria" = "mediumvioletred",
                  "Crustacea" = "indianred1",
                  "Porifera" = "khaki1",
                  "Mollusca" = "darkorange3",
                  "Osteichthyes" = "paleturquoise2",
                  "Echinodermata" = "royalblue4",

```

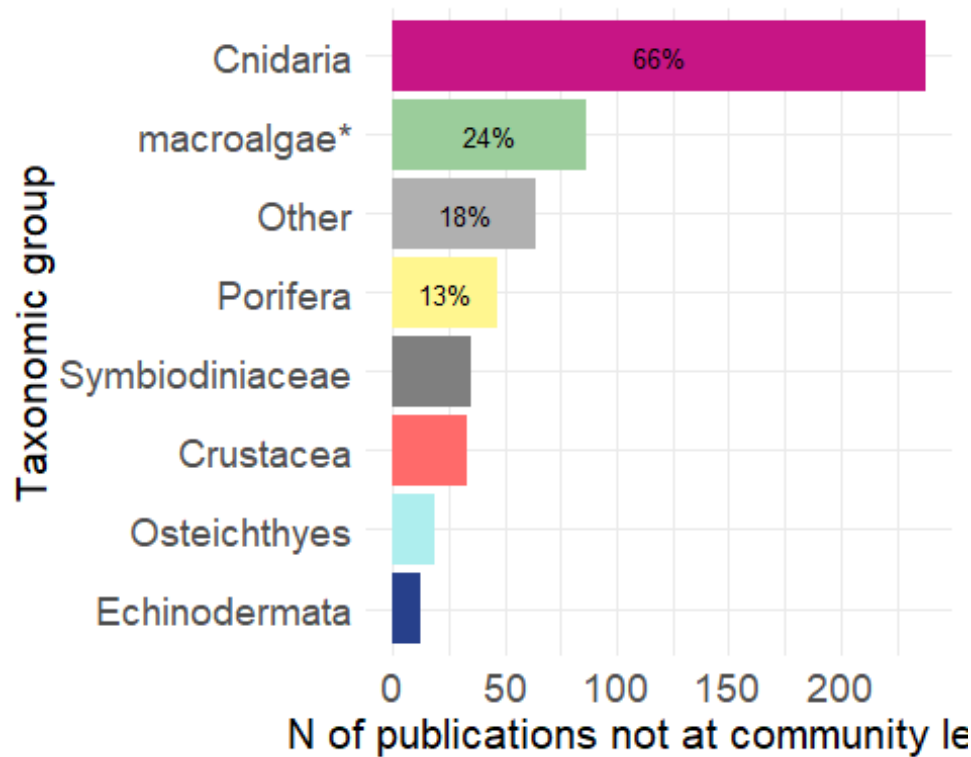
```

    "Tunicata" = "mediumpurple4",
    "Chondrichthyes" = "gray69",
    "Other" = "gray69",
    "Alismatales (seagrasses)" = "gray69",
    "Bryozoa" = "goldenrod1",
    "Annelida" = "burlywood4")

group_bar <- group_bar +
  scale_fill_manual(values = group_palette,
    limits = names(group_palette))+
  theme(axis.text = element_text(size = 14),
    axis.title = element_text(size = 16)
  )

group_bar

```



Export bar plot

```

ggsave("group_bar.png",
  plot = group_bar,
  units = "in", width = 7, height = 6,
  path = here ("Extracted_data_analysis",
    "Output-Visualization"))

```

New species described per phyla

```

new_spec <- tropical %>%
  filter(str_detect(taxonomic_publication, "taxa"))

```

```

new_spec %<>%
  select(File,
         detected_meso_cnidaria,
         detected_meso_sponges,
         detected_meso_seagrass,
         detected_meso_algae,
         detected_meso_mollusca,
         detected_meso_chondrichthyes,
         detected_meso_fish,
         detected_meso_crustacea,
         detected_meso_echino,
         detected_meso_other_taxa)

new_spec_lvls <-
  unique(
    unlist(
      str_split(new_spec$detected_meso_other_taxa, "\\+")
    )
  )
new_spec_lvls<- new_spec_lvls[! new_spec_lvls %in% c("",NA,"notapp")]
new_spec[,new_spec_lvls] <- sapply(new_spec_lvls,
                                   function(y){
                                     str_count(new_spec$detected_meso_othe
r_taxa, y)
                                   }
                                   )

rm(new_spec_lvls)

new_spec %<>%
  mutate(across(everything(), as.character)) %>%
  replace(is.na(.), "0") %>%
  select(-detected_meso_other_taxa) %>%
  filter(if_all(everything(), ~ !grepl("notapp", .x))) %>%
  mutate(across(-File, ~ case_when(
    .x %in% c("x") ~ "1",
    TRUE ~ .x
  ))) %>%
  mutate(across(-File, as.numeric))

new_spec_tot <-
  new_spec %>%
  summarise(across(where(is.numeric),
                    list(sum)
                  )
            )

rm(new_spec)
new_spec_tot <- reshape2::melt(new_spec_tot, id.vars = NULL)

```

```

colnames(new_spec_tot) <- c("Taxa", "Count")

new_spec_tot %<>%
  mutate_at("Taxa",
            stringr::str_replace, "_1", "") %>%
  mutate_at("Taxa",
            stringr::str_replace, "detected_meso_", "")

new_spec_tot %<>%
  filter(Count>0) %>%
  mutate(Percentage= round(Count / sum(Count) * 100,1))

new_spec_plot <- ggplot(new_spec_tot,
                        aes(x = reorder(Taxa, Count), y = Count,
                            fill= Taxa)) +
  geom_bar(stat = "identity") +
  geom_text(

    data = new_spec_tot %>% filter(Percentage > 8),
    aes(label = paste0(round(Percentage, 0), "%")),
    position = position_stack(vjust = 0.5),
    color = "black",
    size = 3.5
  )+
  labs(y = "N of Publications with new species descriptions", x = "Taxonomic
group") +
  theme_minimal() +
  coord_flip()+ # optional: flips for horizontal bars
  theme(legend.position = "none")+
  scale_x_discrete(labels = rev(c("macroalgae*",
                                "Cnidaria",
                                "Crustacea",
                                "Porifera",
                                "Acari",
                                "Symbiodiniaceae",
                                "Mollusca")))
)

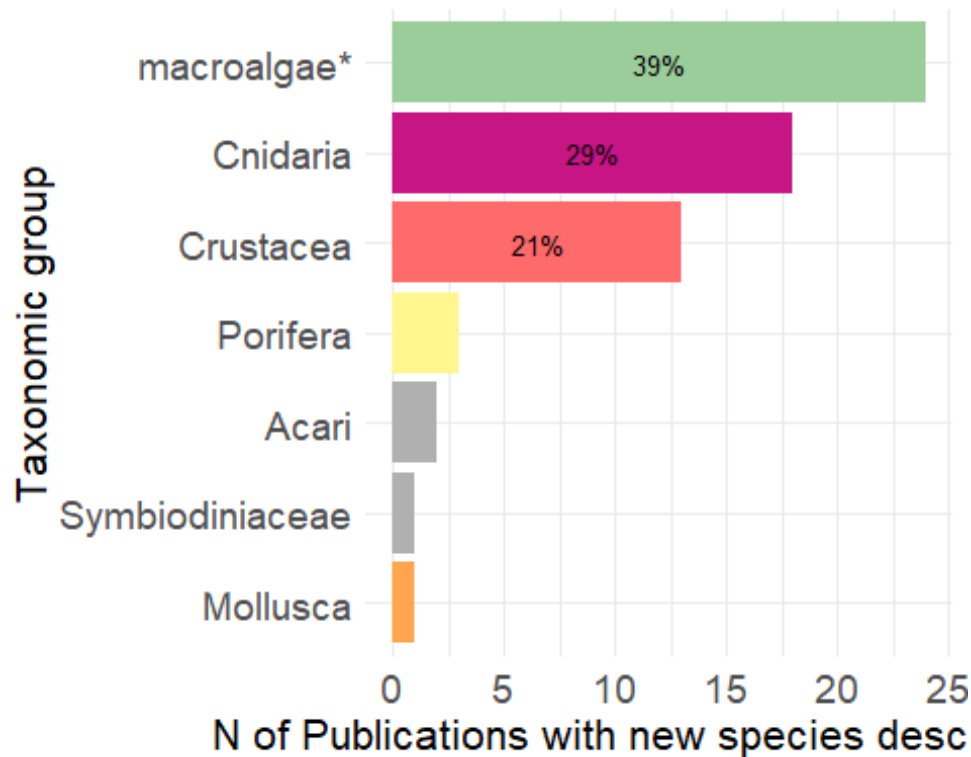
new_spec_palette <- c("algae" = "darkseagreen3",
                     "cnidaria" = "mediumvioletred",
                     "crustacea" = "indianred1",
                     "sponges" = "khaki1",
                     "mollusca" = "tan1",
                     "Acari" = "gray69",
                     "Symbiodiniaceae" = "gray69")

new_species_bar <- new_spec_plot +

```

```
scale_fill_manual(values = new_spec_palette,
  limits = names(new_spec_palette))+
  theme(axis.text = element_text(size = 14),
    axis.title = element_text(size = 16)
  )
```

new_species_bar



Export bar plot

```
ggsave("new_species_bar.png",
  plot = new_species_bar,
  units = "in", width = 7, height = 6,
  path = here ("Extracted_data_analysis", "Output-Visualization"))
```

New species descriptions that included molecular analyses

```
new_spec_molecular <- tropical %>%
  filter(str_detect(taxonomic_publication, "taxa") &
    molecularAnalyses != "no" &
    molecularAnalyses != "nd" &
    molecularAnalyses != "notapp")

nrow(new_spec_molecular)

## [1] 29
```

DATA CROSSING

```
table(tropical$meso_protection)
```

```
##
## mixed      nd      no      yes
##      34     346     14     51
```

VISUALLY SURVEYED SURFACE

```
tropical$visuallySurveyedSurface <- as.factor(tropical$visuallySurveyedSurface)
summary(subset(tropical, is.na(tropical$taxonomic_publication))$visuallySurveyedSurface)
```

##	<1	>1	>10	>100	>1000	>10000	>100000	>1000000
##	1	6	23	50	30	11	9	3
##	nd	notapp						
##	171	80						

Observation

```
print(paste("Number of manipulative experimental studies:", nrow(subset(tropical, grepl("exp", tropical$observationalORexperimental)))))

## [1] "Number of manipulative experimental studies: 66"

print(paste("Number of manipulative experimental studies in the field of restoration:", nrow(subset(tropical, grepl("exp", tropical$observationalORexperimental) & tropical$field_rest != 0))))

)
)
)

## [1] "Number of manipulative experimental studies in the field of restoration: 0"
```

Temp replicates & exp

```
print(paste("# Publications that included temporal replicates:", nrow(subset(tropical, grepl("yes", temporalReplicates)))))

## [1] "# Publications that included temporal replicates: 106"

print(paste("# Publications that included temporal replicates:", nrow(subset(tropical, (grepl("exp", tropical$observationalORexperimental) & grepl("yes", temporalReplicates))))))

## [1] "# Publications that included temporal replicates: 43"
```

Health

```
health <- subset(tropical, grepl("Health", indicators) | grepl("Disease", indicators))
```

```
unique(health$targetTaxa)
```

```
## [1] "Agaricia_grahamae+Agaricia_undata+Madracis_pharensis"
## [2] "Scleractinia"
## [3] "Alveopora_allingi+Alveopora_ocellata"
## [4] "Xestospongia_muta"
## [5] "Amphiprion_chrysopterus"
## [6] "notapp"
## [7] "Scleractinia+Sarcophyton_spp"
## [8] "Antipathes_dichotoma+Antipathes_grandis+Carijoa_riisei"
## [9] "Leptoseris_fragilis+Symbiodiniaceae"
## [10] "Agaricia_lamarcki"
## [11] "Montastraea_cavernosa+Siderastrea_stellata"
## [12] "Stylophora_pistillata+Symbiodiniaceae"
## [13] "Seriatopora_hystrix"
## [14] "Ectocarpus_sp+Clodophora_sp+Lyngbya_sp+Pseudoanabaena_sp"
## [15] "Plakortis_angulospiculatus"
## [16] "Antillogorgia_bipinnata+Ellisella_elongata+Octocorallia+Porifera"
## [17] "Anthozoa"
## [18] "Orbicella_spp"
## [19] "Montastraea_cavernosa+Symbiodiniaceae"
## [20] "Agaricia_lamarcki+Orbicella_franksi"
```

ROV & sample

```
rov <- subset(tropical, grepl("rov", meso_samplingPlatform) )
```

```
rov_specimen <- subset(rovs, grepl("specimen", collectedDataType) )
```

```
print(paste("Percentage of publications relying solely on ROV where specimens  
were collected:", round(nrow(rovs_specimen)/nrow(rovs)*100,1), "%"))
```

```
## [1] "Percentage of publications relying solely on ROV where specimens were  
collected: 60.4 %"
```

```
rm(rovs, rovs_specimen)
```

Molecular techniques in taxonomic and non-taxonomic studies

```
tropical$taxonomic_publication <- as.factor(tropical$taxonomic_publication )
```

```
molecular <- tropical %>%  
  filter(molecularAnalyses != "no" &  
         molecularAnalyses != "notapp" &  
         molecularAnalyses != "nd")
```



```

print(paste("# Publications that adopted molecular techniques: ",nrow(molecular)))

## [1] "# Publications that adopted molecular techniques: 119"

molecular_notaxa <- molecular %>%
  filter(is.na(taxonomic_publication))

print(paste("# Publications that adopted molecular techniques not for taxonomy: ",nrow(molecular_notaxa)))

## [1] "# Publications that adopted molecular techniques not for taxonomy: 90"

rm(molecular, molecular_notaxa)

```

SESSION INFORMATION AND REFERENCES

```

## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
## LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=Italian_Italy.utf8 LC_CTYPE=Italian_Italy.utf8
## [3] LC_MONETARY=Italian_Italy.utf8 LC_NUMERIC=C
## [5] LC_TIME=Italian_Italy.utf8
##
## time zone: Europe/Rome
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] RColorBrewer_1.1-3 treemapify_2.6.0  openxlsx_4.2.8.1  tibble_3.3.1
##
## [5] purrr_1.2.1      stringr_1.6.0      magrittr_2.0.4    readxl_1.4.5
##
## [9] ggplot2_4.0.2     dplyr_1.2.0        here_1.0.2
##
## loaded via a namespace (and not attached):
## [1] utf8_1.2.6        generics_0.1.4     tidyr_1.3.2       stringi_1.8.7
##
## [5] digest_0.6.39     evaluate_1.0.5     grid_4.5.2        fastmap_1.2.0
##
## [9] cellranger_1.1.0  rprojroot_2.1.1    plyr_1.8.9        ggrepel_0.9.6

```

```

## [13] zip_2.3.3          scales_1.4.0      textshaping_1.0.4 cli_3.6.5
## [17] rlang_1.1.7        withr_3.0.2      yaml_2.3.12      otel_0.2.0
## [21] tools_4.5.2        reshape2_1.4.5   forcats_1.0.1    vctrs_0.7.1
## [25] R6_2.6.1           lifecycle_1.0.5  ragg_1.5.0       pkgconfig_2.0.3
## [29] pillar_1.11.1      gtable_0.3.6     glue_1.8.0       Rcpp_1.1.1
## [33] ggfittext_0.10.3   systemfonts_1.3.1 xfun_0.56        tidyselect_1.2.
1
## [37] rstudioapi_0.18.0 knitr_1.51        farver_2.1.2     htmltools_0.5.9
## [41] rmarkdown_2.30     svglite_2.2.2    labeling_0.4.3   compiler_4.5.2
## [45] S7_0.2.1

library(report)
report::report(sessionInfo())

## Analyses were conducted using the R Statistical language (version 4.5.2; R
  Core
## Team, 2025) on Windows 11 x64 (build 26200), using the packages magrittr
## (version 2.0.4; Bache S, Wickham H, 2025), report (version 0.6.3; Makowski
  D et
## al., 2023), here (version 1.0.2; Müller K, 2025), tibble (version 3.3.1; M
  üller
## K, Wickham H, 2026), RColorBrewer (version 1.1.3; Neuwirth E, 2022), openx
  lxx
## (version 4.2.8.1; Schauburger P, Walker A, 2025), ggplot2 (version 4.0.2;
## Wickham H, 2016), stringr (version 1.6.0; Wickham H, 2025), readxl (versio
  n
## 1.4.5; Wickham H, Bryan J, 2025), dplyr (version 1.2.0; Wickham H et al.,
## 2026), purrr (version 1.2.1; Wickham H, Henry L, 2026) and treemapify (ver
  sion
## 2.6.0; Wilkins D, 2025).
##
## References
## -----
## - Bache S, Wickham H (2025). _magrittr: A Forward-Pipe Operator for R_.
## doi:10.32614/CRAN.package.magrittr
## <https://doi.org/10.32614/CRAN.package.magrittr>, R package version 2.0.4,
## <https://CRAN.R-project.org/package=magrittr>.
## - Makowski D, Lüdecke D, Patil I, Thériault R, Ben-Shachar M, Wiernik B
  (2023).
## "Automated Results Reporting as a Practical Tool to Improve Reproducibilit
  y and

```

```

## Methodological Best Practices Adoption." _CRAN_.
## doi:10.32614/CRAN.package.report
## <https://doi.org/10.32614/CRAN.package.report>,
## <https://easystats.github.io/report/>.
## - Müller K (2025). _here: A Simpler Way to Find Your Files_.
## doi:10.32614/CRAN.package.here <https://doi.org/10.32614/CRAN.package.here>, R
## package version 1.0.2, <https://CRAN.R-project.org/package=here>.
## - Müller K, Wickham H (2026). _tibble: Simple Data Frames_.
## doi:10.32614/CRAN.package.tibble
## <https://doi.org/10.32614/CRAN.package.tibble>, R package version 3.3.1,
## <https://CRAN.R-project.org/package=tibble>.
## - Neuwirth E (2022). _RColorBrewer: ColorBrewer Palettes_.
## doi:10.32614/CRAN.package.RColorBrewer
## <https://doi.org/10.32614/CRAN.package.RColorBrewer>, R package version 1.
1-3,
## <https://CRAN.R-project.org/package=RColorBrewer>.
## - R Core Team (2025). _R: A Language and Environment for Statistical
## Computing_. R Foundation for Statistical Computing, Vienna, Austria.
## <https://www.R-project.org/>.
## - Schauburger P, Walker A (2025). _openxlsx: Read, Write and Edit xlsx F
##iles_.
## doi:10.32614/CRAN.package.openxlsx
## <https://doi.org/10.32614/CRAN.package.openxlsx>, R package version 4.2.8.
1,
## <https://CRAN.R-project.org/package=openxlsx>.
## - Wickham H (2016). _ggplot2: Elegant Graphics for Data Analysis_.
## Springer-Verlag New York. ISBN 978-3-319-24277-4,
## <https://ggplot2.tidyverse.org>.
## - Wickham H (2025). _stringr: Simple, Consistent Wrappers for Common Str
##ing
## Operations_. doi:10.32614/CRAN.package.stringr
## <https://doi.org/10.32614/CRAN.package.stringr>, R package version 1.6.0,
## <https://CRAN.R-project.org/package=stringr>.
## - Wickham H, Bryan J (2025). _readxl: Read Excel Files_.
## doi:10.32614/CRAN.package.readxl
## <https://doi.org/10.32614/CRAN.package.readxl>, R package version 1.4.5,
## <https://CRAN.R-project.org/package=readxl>.
## - Wickham H, François R, Henry L, Müller K, Vaughan D (2026). _dplyr: A
##Grammar
## of Data Manipulation_. doi:10.32614/CRAN.package.dplyr
## <https://doi.org/10.32614/CRAN.package.dplyr>, R package version 1.2.0,
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